THEORY OF INFILTRATION

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I. Introduction

A very large fraction of the water falling as rain on the land surfaces of the earth moves through *unsaturated* soil during the subsequent processes of infiltration, drainage, evaporation, and the absorption of soil-water by plant roots. Hydrologists, and their textbooks and handbooks, have tended, nevertheless, to pay relatively little attention to the phenomenon of water movement in unsaturated soils. Most research on this topic has been done by soil physicists, concerned ultimately with agronomic or ecological aspects of hydrology; but their colleagues in engineering hydrology have exhibited an increasing interest in this field in recent years.

The present article deals with the theory of infiltration which is one important outcome of the mathematical-physical approach to the study of water movement in unsaturated soils which has been developed over the past 15 years or so, principally in the U.S., England, and Australia. We shall consider briefly the physical basis for the general formalism of this approach and the limits of its applicability; but we shall be concerned principally with developing the general flow equation (a nonlinear Fokker–Planck equation), with describing methods for its solution, and with presenting the solutions and discussing their physical significance.

We define *infiltration* as the process of the entry into the soil of water made available (under appropriately defined conditions) at its surface. This "surface" may be the natural, more or less horizontal upper surface of the soil; or it may be the bed of a natural or artificial furrow or stream, or the walls of a natural or artificial tunnel or cavity. As we shall see, a fundamental point of departure for the study of infiltration is the study of *absorption*, the particular case of infiltration when the effect of gravity may be neglected, as in horizontal systems, in the early stages of infiltration, and in fine-textured soils in which the influence of moisture gradients dominates that of gravity.

II. Water Transfer in Unsaturated Soils

A. Hydraulic Conductivity of Unsaturated Soils

1. *Darcy's Law for Saturated Media*

We may write Darcy's law [1] for the flow of water in the liquid phase through saturated porous media, including soils, in the form

$$\mathbf{U} = -K \nabla \Phi \tag{1}$$

where $\mathbf{U}$ is the vector flow velocity, Φ is the total potential, and K is the hydraulic conductivity of the medium.

$\mathbf{U}$ has the character of a macroscopic mean. It is a flow rate per unit cross section of the medium formed by averaging over an area large compared with the cross section of individual pores and grains of the medium. $\mathbf{U}$ has the dimensions [length][time]$^{-1}$, and we shall use the specific unit cm-sec^{-1}.

Φ is the potential defined by the equation

$$\Phi = (P/\rho g) + \Omega \tag{2}$$

where P is the pressure (more precisely, the resultant stress due to the hydrostatic and adsorptive force fields), ρ the density of water, g the acceleration due to gravity, and Ω the potential of the external forces per unit weight of water. Like $\mathbf{U}$, P is a macroscopic quantity and is the outcome of smoothing the actual microscopic distribution of pressure in the water over a volume rather larger than that of the individual pores. Usually the only external force on the system is that of gravity, and then $\Omega = z$, the vertical ordinate. As we have defined it, Φ has the dimensions [length]. We shall use the unit centimeters for Φ.

We employ this definition of Φ, which corresponds to the *hydraulic head* used by engineers in their flow calculations, because it is convenient and simple for the purposes of our exposition. The formalism could be developed equally well in terms of a potential based on unit mass. It is an elementary matter to transpose our various results into this form—at the expense of the intrusion of the factor g into various expressions, and the use of a more cumbersome set of units.

It then follows that K has the dimensions [length][time]$^{-1}$, and that we shall specifically use the unit cm-sec^{-1}.

If interaction between medium and water is negligible, K is related to the *intrinsic permeability* of the medium, κ, by the expression

$$K = \kappa \rho g / \mu \tag{3}$$

where μ is the dynamic viscosity, and κ, which depends solely on the internal geometry of the medium, has the dimensions [length]^2.

Equation (1) holds for an isotropic medium, the extension to anisotropic media being

$$\mathbf{U} = -\mathbf{K}\,\nabla\Phi \tag{4}$$

where $\mathbf{K}$ is the hydraulic conductivity tensor.

A great deal has been written in recent years on the theoretical basis of Darcy's law, although much of it seems to constitute self-education rather than serious research. It is sufficient here to remark that there are both hydrodynamic and statistical elements to the basis of Darcy's law. The hydrodynamic one is as follows: the Navier-Stokes equation governing the flow of a viscous, incompressible, Newtonian fluid (e.g., Lamb [2]) is linear in the limit of small Reynolds number. We have already hinted at the statistical one in defining $\mathbf{U}$ and P: it is that, although it is not feasible to know details of the internal geometry of the medium and of the *microscopic* distribution of velocity and potential, local mean quantities such as $\mathbf{U}$ and Φ can be defined and exist.

2. *Darcy's Law for Unsaturated Media*

It has long been assumed [3], and was confirmed experimentally by Childs and Collis-George [4] and others, that Darcy's law may hold for the flow of liquid water in unsaturated media in a modified form in which K is a function of the volumetric moisture content, θ. The theoretical validity of this concept depends on the (usually very reasonable) assumption that the drag at the air-water interfaces in the soil is negligibly small [5]. The general behavior of the $K(\theta)$ function is now fairly well established, thanks to the work of Richards [3], Moore [6], Childs and Collis-George [4], and other workers in soil physics and petroleum engineering research. K is found to decrease very rapidly as θ decreases from its saturation value. This is not surprising, for the following reasons [7]:

(a) The total cross section available for flow decreases with θ.

(b) The largest pores are emptied first as θ decreases. Since the contribution to K per unit area varies roughly as the square of pore "radius," K can be expected to decrease much more rapidly than θ.

(c) As θ decreases, the probability increases that water will occur in pores and wedges isolated from the general three-dimensional network of water films and channels. Once continuity fails, there can be no flow in the liquid phase, other than flow through liquid "islands" in series-parallel with the vapor system [8]. (Flow of this type is usually negligible in the absence of temperature gradients, and need not detain us here.)

K may, in fact, vary through six or more decades over the θ-range of interest [9]. Figure 1 shows a typical $K(\theta)$ relationship.

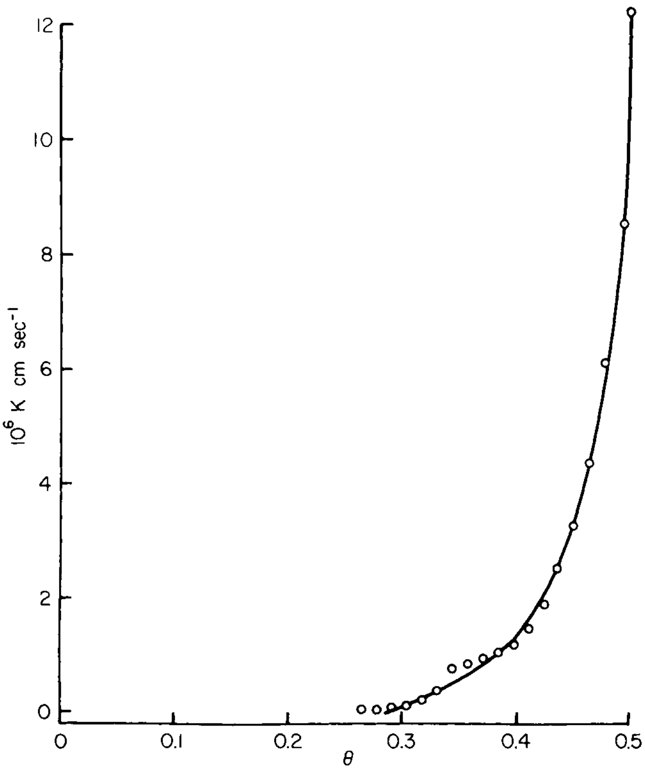

FIG. 1. Relationship between hydraulic conductivity, K, and moisture content, θ, for Yolo light clay [6].

Although the possibility of hysteresis in $K(\theta)$, analogous to that in $\Psi(\theta)$ discussed in Sect. D.6 below, cannot be excluded *a priori*, present indications are that it is relatively unimportant [10–13]. It would not, in any case, enter into the nonhysteretic class of soil-moisture phenomena which is our principal theme here.

We thus express Darcy's law for the flow of water in unsaturated porous media, including soils, in the modified form of Eq. (1):

$$\mathbf{U} = -K(\theta)\, \nabla\Phi \tag{5}$$

B. Moisture Potential and Total Potential of Water in Unsaturated Soils

The water in unsaturated soils is not "free" in the thermodynamic sense, because of capillarity and adsorption, the latter tending to be significant in dry soils [14]. The energy state of the water is commonly expressed by the *moisture potential*, Ψ, which has, as well, been known variously as *capillary potential*, *capillary pressure*, *moisture tension*, *moisture suction*, *negative pressure*, *pressure head*, etc. [15–19]. We here define Ψ as a potential based on unit weight of water, so that, like Φ, it has the dimensions [length] and is conveniently expressed in the unit centimeters. The convenient datum of Ψ is the potential of free water at atmospheric pressure. The appropriate differential specific Gibbs function is then $g\Psi$. It follows that, if the effect of solutes is negligible, the liquid and vapor water systems are connected by the relation

$$h = \exp[g\Psi/\mathbf{RT}] \tag{6}$$

where h is the relative humidity, $\mathbf{R}$ (erg-gm^{-1}-°K^{-1}) is the gas constant for water vapor ($= 4.65 \times 10^6$), and $\mathbf{T}$(°K) is the absolute temperature. It is important to note that, in unsaturated media that are wet by water, Ψ is negative. Figure 2 depicts the $\Psi(\theta)$ relation for the soil for which $K(\theta)$ is given in Fig. 1. Ψ decreases rapidly as θ decreases. When θ is so small that water is retained only in the smallest pores, or in thin adsorbed films, the forces holding it are very great indeed.

For unsaturated media, the total potential Φ may be regarded as comprising Ψ and the gravitational component z, the height above some datum level. Thus, under these conditions, Ψ takes the place of $P/\rho g$ in Eq. (2), Ω is set equal to z, and we have

$$\Phi = \Psi(\theta) + z \tag{7}$$

C. General Partial Differential Equation of Soil-Water Transfer

Combining Eq. (5) with the requirement of continuity that

$$\partial\theta/\partial t = -\nabla \cdot \mathbf{U} \tag{8}$$

we obtain the equation

$$\partial\theta/\partial t = \nabla \cdot (K\, \nabla\Phi) \tag{9}$$

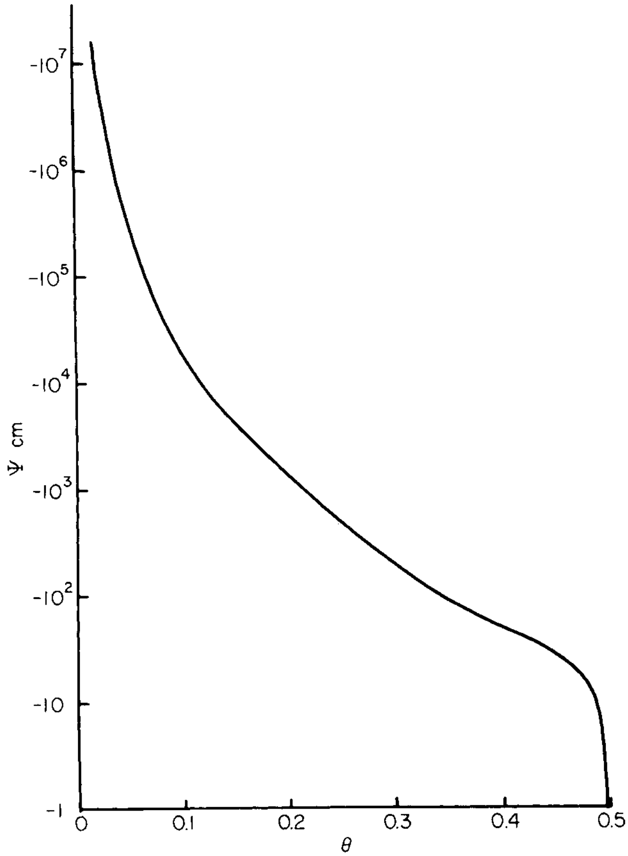

FIG. 2. Relationship between moisture potential, Ψ, and moisture content, θ, for the soil of Fig. 1 [6].

Here t denotes time (sec). Substituting Eq. (7) in Eq. (9), we then have

$$\partial\theta/\partial t = \nabla \cdot (K\,\nabla\Psi) + (\partial K/\partial z) \tag{10}$$

Equations (9) and (10) hold quite generally in the sense that they apply equally well to both homogeneous and heterogeneous soils, and do not depend on any requirement that relations between K, Ψ, and θ be single-valued. For a homogeneous soil, however, we may express Eq. (10) in more tractable form in the two following ways. If K and Ψ are single-valued functions of θ, we may introduce the quantity D, also a single-valued function of θ, such that

$$D = K(d\Psi/d\theta) \tag{11}$$

Then Eq. (10) may be written as

$$\frac{\partial \theta}{\partial t} = \nabla \cdot (D \, \nabla \theta) + \frac{dK}{d\theta} \cdot \frac{\partial \theta}{\partial z} \tag{12}$$

The coefficient $dK/d\theta$ is evidently a function of θ. Alternatively, if K and θ are single-valued functions of Ψ, Ψ may be taken as the dependent variable and Eq. (10) rewritten as

$$\frac{d\theta}{d\Psi} \cdot \frac{\partial \Psi}{\partial t} = \nabla \cdot (K \, \nabla \Psi) + \frac{dK}{d\Psi} \cdot \frac{\partial \Psi}{\partial z} \tag{13}$$

The coefficients $d\theta/d\Psi$ and $dK/d\Psi$ are functions of Ψ.

Contrary to the assertion of a recent review [19], Eqs. (12) and (13) are not completely equivalent. For example, Eq. (13) may still apply when Ψ exceeds the *air-entry value* (i.e., the value at which air enters an initially saturated porous body [20]) or is positive (as it may be when a depth of free water is ponded over the soil), whereas Eq. (12) cannot [21, 22]. In general, however, Eq. (12) is the more tractable form. We are here concerned almost exclusively with solutions of Eq. (12); but it should be understood that, whenever a solution of Eq. (12) can be found, a corresponding solution of Eq. (13) can be found by the same (or a slightly modified) method (see Sects. IV.A, V.A, and V.B).

Equation (12) is a *nonlinear Fokker–Planck equation.* Linear forms of the Fokker–Planck equation arise in such physical phenomena as heat conduction from sources moving relative to the medium [23] and diffusion under an external force field (e.g., sedimentation with Brownian motion) and in the theory of Markov processes in mathematical probability (e.g., Chandrasekhar [24], Bailey [25]). It has entered some recent studies of turbulent diffusion [26, 27]. Nonlinear forms of the Fokker–Planck equation do not seem to have been studied other than in the present context, although work on the closely related nonlinear diffusion equation

$$\partial \theta / \partial t = \nabla \cdot (D \, \nabla \theta) \tag{14}$$

goes back at least to Boltzmann [28]. Equation (12) reduces to Eq. (14) for horizontal systems and in other instances where gravity may be neglected.

It will be understood that Eq. (12) is of the diffusion, or heat-conduction, form but with two complications that add very greatly to the difficulty of its solution. Firstly, the equation contains on its right side the first-order term representing the influence of gravity on the process. Secondly, the coefficients D and $dK/d\theta$ are both markedly dependent on θ. D may vary typically through three or more decades (and $dK/d\theta$ through several more) between the wet and dry ends of the moisture range. Neither of these complications can be ignored, so that the quantitative study of the physics of soil-water transfer depends quite centrally on the availability of solutions to the nonlinear diffusion and Fokker–Planck equations.

D is called the *moisture diffusivity*. It has the dimensions [length]^2 [time]$^{-1}$, and the unit cm^2-sec^{-1} is appropriate. Figure 3 is a graph of $D(\theta)$ for the soil with the $K(\theta)$ and $\Psi(\theta)$ functions of Figs. 1 and 2.

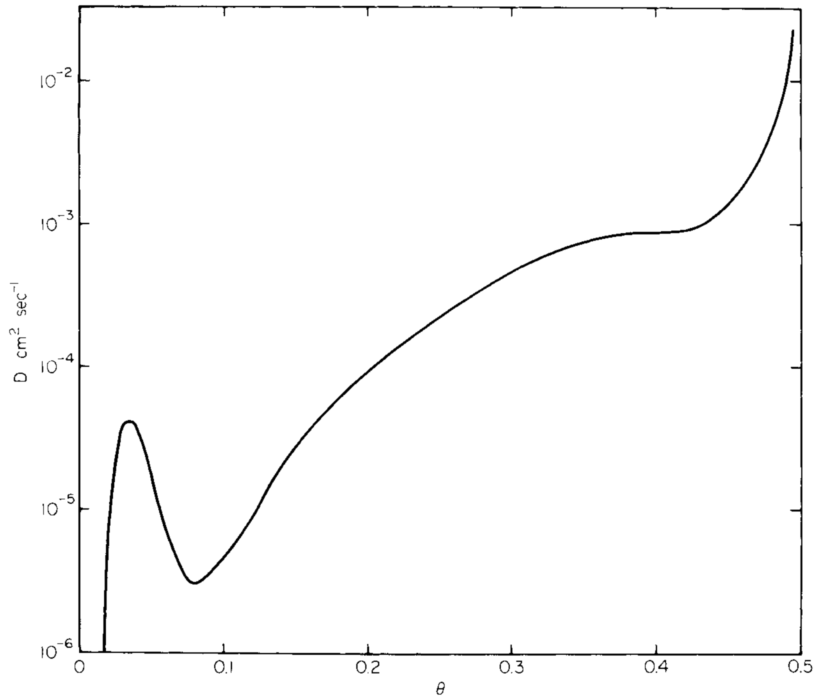

FIG. 3. Relationship between moisture diffusivity, D, and moisture content, θ, for the soil of Fig. 1 [21]. (Note that D for $\theta \leq 0.06$ includes dominant contribution in vapor phase.)

Many of the concepts leading to Eq. (12) were implicit in the 1907 monograph of Buckingham [29]. In 1931, Richards [3] presented equations formally equivalent to Eqs. (10) and (13). In 1948, Childs and George [30] recognized the diffusion character of the particular form of Eq. (14) which applies to a one-dimensional horizontal system. In 1952, Klute [31, 32] explicitly derived Eq. (12). Philip [21, 33, 34] extended the approach to include moisture transfer in the vapor and adsorbed phases within the same mathematical formalism. These extensions depended, essentially, on the use of Eq. (6) and the chain rule for differentiation.

D. Limits to Applicability of Approach

The remainder of this article is, for the most part, concerned with the development of solutions of Eq. (12) which are relevant to the process of infiltration, and with the physical implications of these solutions. In a sense, our study can be looked on as an essay on infiltration as the outcome of physical interactions between capillarity, gravity, and geometry. (Here, and in what follows, the term *capillarity* is convenient to describe the consequence of gradients of Ψ, although we recognize that the principal determinant of Ψ in very dry soils and in soils of high colloid content may not be the curvature of air-water interfaces.)

It is, however, desirable that we discuss the limits to the applicability of the approach, not only so that the reader will not overestimate it, but equally so that he will not underrate it (as he might well do from a superficial reading of some of the literature). We therefore present here a very brief but, it is hoped, fairly inclusive review of the ways in which deviations from the formalism of Sect. II.C may arise.

It is convenient to develop the discussion under seven headings, although it will be understood that the complications of the subject are such that these are not all mutually exclusive.

1. Inadequacy of Specification on Darcy Scale. Local Disequilibrium in Aggregated Media

The possibility must be considered that the specification of the medium "on the Darcy scale" (i.e., in terms of quantities such as K, Ψ, and θ based on volume elements large compared with that of the individual pore) may not be adequate. Where the medium is aggregated, i.e., where it contains aggregates microporous that form the "grains" of a macroporous (or a "cracked") structure, it must be admitted *a priori* that, during transient phenomena, local disequilibria between Ψ values in the microporosity and in the macroporosity may invalidate any description of the process at the Darcy level. A recent detailed analysis of the problem [35, 36], however, indicates that deviations produced in this way will usually be negligible; for the exceptional case of very large aggregates of low sorptivity (see Sect. IV for a definition of this term), the approaches developed by Philip [35, 36] provide the necessary extensions of the methods of analysis discussed in this article.

2. Deviations from Darcy's Law

We discuss two possible causes of deviations from Darcy's law.

a. INERTIAL EFFECTS. As we have already remarked, the Navier–Stokes equation underlying Darcy's law ceases to be linear when the "inertia terms" are important. For steady flows in porous media, this has the consequence that Darcy's law fails when the Reynolds number of the flow (based on $|\mathbf{U}|$ and on a characteristic pore dimension) exceeds a value that tends to lie in the range 1 to 10. Some authors have mistakenly ascribed deviations from proportionality between $\mathbf{U}$ and $\nabla\Phi$ to turbulence, although it is definitely a nonlinear laminar regime that emerges as the Reynolds number increases, and not a turbulent one [37, 38]. The Reynolds numbers characteristic of steady soil-water flows are very small indeed, and the possibility of the failure of Darcy's law due to inertial effects cannot be entertained in such cases.

The influences of inertia on transient flows are somewhat different, and the matter requires more careful examination. Analyses by Philip, however, indicate that the deviations from Darcy's law due to inertia are unlikely to be of much significance during transient flows in either saturated [39] or unsaturated [40] soils.

b. NON-NEWTONIAN BEHAVIOR. The applicability of the Navier–Stokes equation depends also on the soil-water behaving as a *Newtonian fluid* (in which the stresses are proportional to the rates of strain). *Bulk water* is a Newtonian fluid, and it would seem that the flow behavior of soil-water can differ significantly from that of bulk water only if a large fraction of the soil-water is subject to surface forces capable of modifying its rheological properties in the required way. Deviations from Darcy's law due to non-Newtonian behavior may therefore be possible in principle in soils of high colloid content and large specific surface; and observations of "non-Darcy" behavior tend to be on soils of this type or, indeed, on pastes made of water and colloidal extracts. The work of Olsen [41] suggests that at least some of these observations are the spurious outcome of poor experimental technique. Swartzendruber [42] has considered various physical explanations for these "non-Darcy" observations and favors the non-Newtonian explanation.

An aspect arises in connection with unsaturated systems which does not seem to have been discussed (although it is perhaps suggested by Olsen's paper). This concerns the influence of the *surface viscosity* of air-water interfaces, and of the abnormal rheology of such surfaces, which are, of course, natural regions of accumulation of contaminants.

These ideas lead us into areas of surface physical chemistry well beyond the scope of this article. We mention them only to indicate that the question of the reality or otherwise of non-Newtonian behavior seems to be a matter for rheologists and surface physical chemists. We shall not allow it to detain us here, as it does not appear to have much bearing on the movement of soil-water in general, or, in particular, on the process of infiltration.

3. *Other Complications due to Colloidal Behavior*

a. COLLOID SWELLING. The processes of swelling and shrinkage in soils of high colloid content involve the motion of the soil particles. The movement of water in one-dimensional systems of this type may be analyzed through the equation obtained by combining the continuity requirement and Darcy's law applied (as it should be) to flow *relative to the soil particles.* Appropriate methods of developing the physical theory, which are similar in principle to those to be discussed in this article, are presently under active development by Dr. D. E. Smiles and the author. This work, which is too recent for an adequate account of it to be possible in this article, points up the importance of mass flow (i.e., of water convected with the soil particles) in such systems.

b. EFFECTS OF ELECTROLYTE CONCENTRATION. The variation of the structure and hydraulic conductivity of soils of high colloid content with electrolyte concentration [43] is, of course, not specifically taken into account in the present study.

4. *Possible Complications due to Soil-Air*

The analysis developed in Sect. II.C neglects any effect of pressure differences in the soil-air. We may develop a set of equations similar to the foregoing, which embrace flows of both soil-water and soil-air. This, slightly more general, two-component formalism is, in fact, used in petroleum engineering. In most applications to soil-water movement, this elaboration is unnecessary, as the pressure differences within the soil-atmosphere are trivially small. We discuss below certain instances where this is not so, and certain other complications that may arise from interactions between the soil-air and the soil-water.

a. AIR ESCAPE IMPEDED OR PREVENTED. The dynamics of processes such as the absorption and infiltration of water may be modified significantly if the air initially in the soil is not free to escape as it is displaced by water. The process may be studied, to some extent, by use of the two-component formalism mentioned previously (see Elrick [44] for a linearized attack on a problem of this type). As Youngs and Peck have pointed out [45], however, capillary hysteresis may lead to unexpected complications. Also, as the experiments of Peck [46, 47] indicate, air pressures may increase to the point where the pressure buildup is partly relieved by a bubbling air escape at the water-supply surface. Such a phenomenon is, of course, beyond the scope of a Darcy-type analysis.

These considerations do not appear to be generally of much importance in the field, although, as this author is well aware from his personal experiences in the Riverina of Australia, limits to air escape may well affect infiltration into large inundated areas. In fact, soil-air pressures have been developed which are great enough to lift the pavements of highways passing through the flooded region.

b. ENTRAPPED AIR. As Philip [48] pointed out, air is more likely to be trapped in pores remote from the water-supply surface than in those near the surface. Such spatial differences in air-entrapment will be reflected in spatial differences in the $\Psi(\theta)$ relation. The effect is thus essentially one of spatial heterogeneity (Sect. IX), but it is apparently of minor significance. It is a possible explanation of the sometimes observed *transition zone* in moisture profiles during infiltration (Sect. X).

c. THE SOLUTION OF SOIL-AIR. Relatively little consideration appears to have been given to the diffusion of air dissolved in the soil-water, and to the associated processes of air going into, and coming out of, solution. Such processes clearly lead to changes in the incidence of entrapped air, and may thus affect the stability of the $K(\theta)$ and $\Psi(\theta)$ relations in the medium. It has been remarked elsewhere [48] that the use of air-free water, often a (supposed) precaution in laboratory experiments, leads to the gradual solution of trapped air, with consequent changes in medium properties. It seems preferable to use water in equilibrium with the atmosphere, since the experimental conditions are then closer to nature, and the complications of progressive solution of entrapped air are avoided. Peck [49] has suggested that the use of initially de-aired water may be the cause of the deviant behavior in absorption experiments by Nielsen *et al.* [50].

5. *Thermal Effects*

The analysis of Sect. II.C refers, as it stands, to isothermal systems. The extensions to nonisothermal systems have been made by Philip and de Vries [8, 21, 51–53]; analysis of such systems involves, in general, solution of simultaneous equations for both heat and moisture transfer. Recent attempts to study these matters through the formalism of the thermodynamics of irreversible processes do not seem to have yielded any real progress. The principal specific result [54] seems to reduce to the identification of the well-known proportionality of water vapor and latent heat transfer in an evaporation-condensation system as an Onsager reciprocity relation.

It has been shown (e.g., Philip and de Vries [8, 51]) that thermally induced moisture movement tends to be most important in rather dry systems where vapor diffusion is a dominant mechanism. The processes of absorption and infiltration treated in this article take place almost exclusively in the liquid phase, and we may safely limit our attention to isothermal systems.

The possibility that thermal effects arising from the heat of wetting might have a significant influence on processes such as absorption and infiltration has recently been studied experimentally [55, 56]. The authors concluded that such effects are quite negligible even in soils initially so dry that Ψ is significantly less than $-15{,}000$ cm. A similar conclusion was reached in an earlier, theoretical investigation of the energetics of absorption and infiltration [57, 58].

6. *Capillary Hysteresis*

Perhaps the most important and challenging limitation to the applicability of Eqs. (12) and (13) arises from the fact of hysteresis in the $\Psi(\theta)$ relation, first discussed by Haines [59]. As Philip [21] remarked, hysteresis does not in itself invalidate the use of Eqs. (12) and (13), so long as the $\Psi(\theta)$ relation adopted is appropriate to the phenomenon under study. A wetting curve is thus the $\Psi(\theta)$ function appropriate to the analysis of absorption or infiltration into a homogeneous soil of uniform initial moisture content; and, conversely, a drying curve should be used in the analysis of suitably uniform processes of removal of soil-water. However, as Childs first pointed out [21], it is not, in general, useful to treat in this way phenomena, such as redistribution and drainage after infiltration ceases, wherein wetting and drying occur simultaneously in the one soil mass. In general, the "diffusion" analysis fails in such cases: different points in the soil mass follow different hysteresis scanning curves, so that there is no definite relationship between gradients of Ψ and gradients of θ. In such cases, a knowledge of the complete family of scanning curves within the main $\Psi(\theta)$ hysteresis loop is prerequisite to any quantitative treatment of the transfer process. (Some special cases of simultaneous wetting and drying to which the diffusion analysis is, nevertheless, relevant have been treated by Philip [7] and Youngs [60].)

Collis-George [61] suggested that the independent domain model of hysteresis due to Everett and others [62–64] might be applicable to capillary hysteresis. Miller and Miller [65] gave an original and perceptive treatment of capillary hysteresis, but not a quantitative one. Poulovassilis [66, 67] gave the first published account of capillary hysteresis in terms of the independent domain model and presented experimental evidence for its validity. Poulovassilis used an improved form of the model [68–70]. As Philip [71, 72] showed, the approach is most simply described without any specific reference to independent domain theory. It depends solely on the supposition that any infinitesimal volume element of the pore space may be characterized by two values of Ψ: α, the (arithmetically) smallest value at which, under equilibrium conditions, the element may be occupied by air; and β, the (arithmetically) largest value at which the element may, under equilibrium conditions, be occupied by water. If, then, each volume element of the total pore space is characterized by its α and β values, there exists a bivariate distribution density function $f(\alpha, \beta)$ such that the fraction

$$f(\alpha_0, \beta_0)\, d\alpha\, d\beta$$

of the total pore volume consists of elements for which α lies between the values α_0, $\alpha_0 + d\alpha$, and β lies between the values β_0, $\beta_0 + d\beta$. Then $f(\alpha, \beta)$ completely defines the hysteretic behavior of the medium. Philip [72] showed that a similarity hypothesis about $f(\alpha, \beta)$ (which implied, loosely, that the distribution of geometrical relationships between "wetting" and "drying" air-water interfaces is independent of pore size) holds fairly well for Poulovassilis' data, and that the use of this hypothesis enables $f(\alpha, \beta)$ to be estimated from the main hysteresis loop, without the laborious mapping of scanning curves. Topp and Miller [13] consider that their experiments do not support the independent domain model. Should it prove that the independent domain model is not reasonably adequate, this will seemingly imply that a more complicated formalism is needed. This would not augur well for the prospects of developing a manageable quantitative analysis of hysteretic soil-water processes.

Fortunately, as we have observed previously, capillary hysteresis does not enter the class of phenomena we consider in the following sections. We have discussed the matter at this length because it is a fascinating aspect of the general problem of soil-water behavior, and one that is currently exciting the interest of soil physicists in various parts of the world.

7. *Discussion*

The theory we develop in the succeeding parts of this article represents the *basic fluid-mechanical theory of infiltration.* Those of the complications we have raised, but have not been able to dismiss out of hand as negligible, will, in some circumstances, impose more or less severe perturbations on the basic behavior that the theory predicts. Our theory, nevertheless, provides a natural point of departure for the study of any such perturbations; and it seems clear that it must remain a central and integral part of any more elaborate formalism that aims to take specific account of one or more of these complications.

III. Mathematical Preliminaries

A. Particular Forms of Eq. (12)

In considering specific infiltration phenomena, we shall make use of various particular forms of Eq. (12). It is useful to set out these forms here. There is, of course, an analogous set of specific forms of Eq. (13).

1. Absorption Equations

We have remarked already that Eq. (12) reduces to the nonlinear diffusion equation

$$\partial\theta/\partial t = \nabla \cdot (D\,\nabla\theta) \tag{14}$$

when gravity may be neglected. This then is the equation describing absorption, i.e., infiltration into horizontal systems, or into fine-textured soils in which the influence of the moisture gradients is much more important than that of gravity. As we shall see, absorption solutions have the additional theoretical importance that they yield the limiting small-time behavior of transient infiltration processes even when gravity cannot be neglected, and so provide a basic point of departure for the solution of the (more complicated) infiltration equations.

a. ONE-DIMENSIONAL FORM. The one-dimensional form of Eq. (14) is

$$\frac{\partial\theta}{\partial t} = \frac{\partial}{\partial x}\left(D\,\frac{\partial\theta}{\partial x}\right) \tag{15}$$

with x the space coordinate.

b. TWO-DIMENSIONAL FORMS. The Cartesian two-dimensional form of Eq. (14) is

$$\frac{\partial\theta}{\partial t} = \frac{\partial}{\partial x}\left(D\,\frac{\partial\theta}{\partial x}\right) + \frac{\partial}{\partial y}\left(D\,\frac{\partial\theta}{\partial y}\right) \tag{16}$$

where x, y are the two (Cartesian) space coordinates.

In what follows, we shall be specially concerned with two-dimensional systems exhibiting cylindrical radial symmetry. In such cases, the appropriate form of the equation is

$$\frac{\partial\theta}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(Dr\,\frac{\partial\theta}{\partial r}\right) \tag{17}$$

where r is the radial coordinate.

c. THREE-DIMENSIONAL FORMS. The Cartesian three-dimensional form of Eq. (14) is

$$\frac{\partial\theta}{\partial t} = \frac{\partial}{\partial x}\left(D\,\frac{\partial\theta}{\partial x}\right) + \frac{\partial}{\partial y}\left(D\,\frac{\partial\theta}{\partial y}\right) + \frac{\partial}{\partial z}\left(D\,\frac{\partial\theta}{\partial z}\right) \tag{18}$$

For three-dimensional systems exhibiting spherical radial symmetry, the equation reduces to

$$\frac{\partial\theta}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(Dr^2\,\frac{\partial\theta}{\partial r}\right) \tag{19}$$

d. m-DIMENSIONAL RADIALLY SYMMETRICAL FORM. Equations (15), (17), and (19) are all of the form

$$\frac{\partial \theta}{\partial t} = r^{1-m} \frac{\partial}{\partial r}\left(D r^{m-1} \frac{\partial \theta}{\partial r}\right) \tag{20}$$

where m is the number of dimensions of the system. This form is useful in Sect. IV.

2. *Infiltration Equations*

It turns out that the calculation of solutions of the infiltration equation may be made more naturally and a little more conveniently if we adopt the convention that z is taken *positive downward.* In deference to this convention, we rewrite Eq. (12) as

$$\frac{\partial \theta}{\partial t} = \nabla \cdot (D\, \nabla \theta) - \frac{dK}{d\theta} \cdot \frac{\partial \theta}{\partial z} \tag{21}$$

a. ONE-DIMENSIONAL FORM. The one-dimensional form of Eq. (21) is

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z}\left(D \frac{\partial \theta}{\partial z}\right) - \frac{dK}{d\theta} \cdot \frac{\partial \theta}{\partial z} \tag{22}$$

b. TWO-DIMENSIONAL FORM. The two-dimensional form is

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial x}\left(D \frac{\partial \theta}{\partial x}\right) + \frac{\partial}{\partial z}\left(D \frac{\partial \theta}{\partial z}\right) - \frac{dK}{d\theta} \cdot \frac{\partial \theta}{\partial z} \tag{23}$$

with x the horizontal coordinate.

c. THREE-DIMENSIONAL FORMS. The general Cartesian three-dimensional form is

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial x}\left(D \frac{\partial \theta}{\partial x}\right) + \frac{\partial}{\partial y}\left(D \frac{\partial \theta}{\partial y}\right) + \frac{\partial}{\partial z}\left(D \frac{\partial \theta}{\partial z}\right) - \frac{dK}{d\theta} \cdot \frac{\partial \theta}{\partial z} \tag{24}$$

with x, y the horizontal coordinates. For three-dimensional systems exhibiting cylindrical radial symmetry about a vertical axis, this becomes

$$\frac{\partial \theta}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(D r \frac{\partial \theta}{\partial r}\right) + \frac{\partial}{\partial z}\left(D \frac{\partial \theta}{\partial z}\right) - \frac{dK}{d\theta} \cdot \frac{\partial \theta}{\partial z} \tag{25}$$

where the horizontal radial coordinate $r = (x^2 + y^2)^{1/2}$.

B. Initial and Boundary Conditions

1. Initial Conditions

We shall be concerned almost exclusively with infiltration into a soil mass at uniform initial moisture content θ_0. In such cases, the initial condition governing the appropriate form of Eq. (12) is

$$t = 0, \qquad \theta = \theta_0 \qquad \text{in all} \quad \mathbf{V} \tag{26}$$

where **V** signifies the whole volume of the soil mass. This type of initial condition is a simple and natural one, but some care is needed in its application to the infiltration equation. If θ_0 is not so small that $K_0[= K(\theta_0)]$ is zero, or at least negligibly small, there will be an initial, gravity-caused flow throughout the system and, in infinite systems, a "flow at infinity" throughout the course of the phenomenon. Once this is recognized, it produces no real difficulty beyond a slight complication in certain calculations [34, 58, 73].

2. Boundary Conditions at the Water-Supply Surface

a. CONCENTRATION CONDITIONS. Most of the following analysis deals with the process of absorption or infiltration which follows the continuous supply of water from time $t = 0$ onward to surface **W** at moisture potential Ψ_1. The boundary condition at the water-supply surface is then

$$t \geqq 0, \qquad \Psi = \Psi_1 \qquad \text{at} \quad \mathbf{W} \tag{27}$$

When Eq. (12) is used (as in most of the present article), it is necessary to rewrite this in the form

$$t \geqq 0, \qquad \theta = \theta_1 \qquad \text{at} \quad \mathbf{W} \tag{28}$$

where $\theta_1 = \theta(\Psi_1)$.

Commonly, the water supplied is "free" and under little or no hydrostatic pressure. We may then take $\Psi_1 = 0$, and θ_1 is then the "saturated" moisture content (which may be rather less than the moisture content with all pores water-filled, because of air-entrapment). Where the water depth over the supply surface (or the hydrostatic pressure under which water is supplied) is not negligibly small, and also where the *air-entry value* of the soil is (arithmetically) large, Eq. (13) is the appropriate equation and (27) the appropriate form of the condition at **W**.

In keeping with terminology in the mathematics of diffusion, we designate this type of boundary condition a *concentration condition.*

b. FLUX CONDITIONS. For a second class of infiltration phenomena, water reaches the supply surface at a definite rate (as in infiltration from rainfall, or from sprinkler irrigation). If the soil is able to take up the arriving water for some period $0 \leqq t < t_0$, the appropriate boundary conditions (for one-dimensional systems) are as follows.

For **W** a horizontal surface,

$$0 \leqq t < t_0, \qquad K\,\partial\Psi/\partial z \quad \text{or} \quad D\,\partial\theta/\partial z = K - U_0 \qquad \text{at} \quad \mathbf{W} \tag{29}$$

For **W** a vertical surface,

$$0 \leqq t < t_0, \qquad K\,\partial\Psi/\partial x \quad \text{or} \quad D\,\partial\theta/\partial x = -\,U_0 \qquad \text{at} \quad \mathbf{W} \tag{30}$$

x is here the ordinate normal to **W**. U_0 is the supply rate per unit area of **W**. The first form of the left side of the equalities is appropriate to equations with Ψ the dependent variable and the second to ones with θ the dependent variable.

We refer to conditions like (29) and (30) as *flux conditions*, in accordance with diffusion terminology.

In some circumstances, $t_0 = \infty$, i.e., a condition like (29) and (30) holds for all $t \geqq 0$; but, when t_0 is finite, a second condition at **W** is needed for $t \geqq t_0$. Its form depends on whether the excess water at **W** ponds, runs off, etc.

C. Relevance to Hydrologic Processes Other than Absorption and Infiltration

Although the methods and solutions discussed below are presented specifically in connection with absorption and infiltration, the reader will understand that, with appropriate redefinitions of symbols (and, in some cases, changes of sign), certain of them are relevant also to other processes, such as desorption, drainage, capillary rise, and evaporation from soil [34, 51, 74].

D. Techniques for Solving Nonlinear Diffusion and Fokker–Planck Equations

1. Computer Solutions

The various nonlinear flow equations we have discussed, subject to the conditions we have considered, may, of course, be solved by the brute-force use of finite-difference methods on high-speed computers. There is no special obstacle to using such methods, even for much more elaborate flow problems with one or more of the complications of spatial heterogeneity, more involved initial and boundary conditions, and the operation of capillary hysteresis.

A computer attack on problems of this class will always be feasible provided (i) the problem is well-posed in the mathematical sense; (ii) the (well-established) criteria for computational stability are satisfied by the finite-difference scheme adopted. It should be added that the efficiency of the technique demands that the calculator avail himself of analytical methods to establish the behavior of his solutions in the neighborhood of any singularities.

The ready availability of computer solutions is not, however, so fruitful a potential source of understanding of the theory of infiltration as superficial consideration might suggest. The difficulty is that, in general, each of these nonlinear problems requires for each soil [with its particular hydrological characteristics such as $K(\theta)$ and $\Psi(\theta)$] its own *ad hoc* solution, so that the approach is necessarily a piecemeal one. The possibilities of developing general statements about the phenomena on the basis of calculations of this type are not very good, because they tell us relatively little about the *fundamental structure of the solutions*; and it is upon this which the prospects of useful generalization depend. (It is not my intention to denigrate the use of computers for soil-water calculations; but their most effective rôle in this context seems to me to be as the immensely helpful tools of *ad hoc* investigations.) The emphasis in this article is therefore on quasi-analytical and analytical methods of solution.

2. *Quasi-Analytical and Analytical Solutions*

We shall describe as *quasi-analytical solutions* those in which we may use the methods of mathematical analysis to establish the basic form of the solution, even though some "coefficients" that appear in the solution require to be determined by numerical methods (conceivably with the aid of computers!). Most of the "exact" solutions we shall discuss are of this type.

In some special cases, the solutions may be found completely by mathematical analysis; we shall call such solutions *analytical solutions*.

It is convenient to describe both quasi-analytical and analytical solutions as *exact solutions*. This usage is well established in fluid mechanics [75], although it differs from that of Philip [76].

3. *Analytically Based Estimates of Integral Properties of Solutions*

We shall have occasion to make use of two special classes of analytical solutions, *linearized solutions* and *delta-function solutions*. In particular, we use integral forms of these solutions and a comparison technique (Sect. VIII) to establish estimates of integral properties of certain solutions.

IV. Exact Solutions of Absorption Equations

A. One-Dimensional Quasi-Analytical Solution

Absorption into an effectively semi-infinite, homogeneous, one-dimensional system of uniform initial moisture content is described by Eq. (31) subject to conditions (32):

$$\frac{\partial\theta}{\partial t} = \frac{\partial}{\partial x}\left(D\frac{\partial\theta}{\partial x}\right) \qquad \text{(31) [also Eq. (15)]}$$

$$\begin{aligned} t &= 0, \qquad x > 0, \qquad \theta = \theta_0 \\ t &\geqq 0, \qquad x = 0, \qquad \theta = \theta_1 \end{aligned} \qquad (32)$$

The similarity substitution

$$\phi = xt^{-1/2} \qquad (33)$$

apparently first used by Boltzmann [28], reduces Eqs. (31) and (32) to Eqs. (34) and (35):

$$-\frac{\phi}{2}\frac{d\theta}{d\phi} = \frac{d}{d\phi}\left(D\frac{d\theta}{d\phi}\right) \qquad (34)$$

$$\begin{aligned} \phi &= 0, \qquad \theta = \theta_1 \\ \phi &\to \infty, \qquad \theta = \theta_0 \end{aligned} \qquad (35)$$

The solution of Eq. (31) subject to conditions (32) is therefore

$$x(\theta, t) = \phi(\theta) \cdot t^{1/2} \qquad (36)$$

where $\phi(\theta)$ is the solution of Eq. (34) subject to conditions (35).

Klute [31, 32] used a numerical method due to Crank and Henry [77, 78] to find $\phi(\theta)$. Philip [33, 34, 79] subsequently developed a much more rapid and accurate method. Equation (34) may be rewritten as

$$-\frac{\phi}{2} = \frac{d}{d\theta}\left(D\frac{d\theta}{d\phi}\right)$$

and integration yields

$$\int_{\theta_0}^{\theta} \phi\, d\theta = -2D\, d\theta/d\phi \qquad (37)$$

subject to the condition

$$\theta = \theta_1, \qquad \phi = 0 \qquad (38)$$

The lower limit of the integral is θ_0 because the second of conditions (35) implies $d\theta/d\phi = 0$ as $\theta \to \theta_0$.

Philip's method uses an appropriate finite difference form of Eq. (37) in an iterative procedure that is initiated by an estimate of $\int_{\theta_0}^{\theta_1} \phi \, d\theta$. A prescription for making the first estimate is given by Philip [79]. The method employs the Richardson h^2-extrapolation [80] to reduce truncation errors, and avoids difficulties connected with the infinite tail (which occurs as $\theta \to \theta_0$ unless $D(\theta_0) \to 0$ sufficiently rapidly [48]) by using an analytical solution in that region.

The method applies in slightly modified form [81] when water is supplied under positive hydrostatic pressure, and Eq. (31) must be replaced by the relevant form of Eq. (13), at least in the region of Ψ positive.

B. One-Dimensional Analytical Solution

Subsequently, Philip [76, 82, 83] showed that there exists an indefinitely large class of functional forms of $D(\theta)$ for which $\phi(\theta)$ in Eq. (36) [and therefore the solution of Eq. (31) subject to conditions (32)] may be found analytically. Equation (37) may be rewritten as

$$D = -\tfrac{1}{2}\, d\phi/d\theta \cdot \int_{\theta_0}^{\theta} \phi \, d\theta \tag{39}$$

It follows that any functional form of ϕ,

$$\phi = F(\theta) \tag{40}$$

which satisfies the simple conditions we state below, must be the solution of Eq. (34) subject to conditions (35) for the case where

$$D(\theta) = -\tfrac{1}{2}\, dF/d\theta \cdot \int_{\theta_0}^{\theta} F \, d\theta \tag{41}$$

D is evidently expressible in terms of known functions if F is integrable in terms of known functions. F must satisfy three conditions:

(i) The condition imposed by (35) is that $F(\theta_1) = 0$.

(ii) The condition imposed by the requirement that $D(\theta)$ exists for $\theta_0 \leqq \theta \leqq \theta_1$ is that both $\int_{\theta_0}^{\theta} F \, d\theta$ and $dF/d\theta$ exist throughout $\theta_0 < \theta \leqq \theta_1$. (But, if a finite number of discontinuities in D are allowable, or if D is permitted to be infinite at a finite number of points in the range $\theta_0 < \theta \leqq \theta_1$, $dF/d\theta$ may either not exist or be infinite at the appropriate number of points in the θ range.)

(iii) The condition imposed by the nonnegativity of D is that $dF/d\theta$ be nonpositive in the range $\theta_0 \leqq \theta \leqq \theta_1$.

Some restriction to the utility of this method is imposed by the result that $D(\theta_0)$ is finite, nonzero, only if

$$\lim_{\theta \to \theta_0} F(\theta)/[-\log(\theta - \theta_0)]^{1/2}$$

is finite, nonzero. The most readily generated *D-F* analytical pairs satisfying this criterion are rather complicated in form, and are not especially well adapted to fitting empirical data on $D(\theta)$. More careful examination of the whole class of F functions, with special reference to those that behave like $[-\log(\theta - \theta_0)]^{1/2}$ as $\theta \to \theta_0$, is needed before the potentialities of this approach will have been explored and exploited.

C. The Sorptivity of Porous Media and Soils

The one-dimensional absorption result discussed in Sects. IV.A and IV.B is the simplest transient solution of the nonlinear Fokker–Planck equation. As will be seen from the succeeding developments here, it plays a central rôle in the study of more difficult problems involving geometrical complications and the effect of gravity. As we shall show, the early stages of the infiltration process are virtually independent of both gravity and geometry, and are essentially the same as one-dimensional absorption: it follows that the preceding solution represents the leading term of perturbation-type analyses of more complicated problems. It is for this reason that the sorptivity, S, which arises naturally in the one-dimensional absorption problem and which we define below, is so important in the development of the theory of infiltration.

1. Sorptivity

Reverting to the solution (36), we introduce the symbol i (dimension [length]; convenient unit centimeters) to designate the cumulative absorption into the medium at time t. It follows at once that

$$i = \int_{\theta_0}^{\theta_1} \phi \, d\theta \cdot t^{1/2}$$

i.e.,

$$i = St^{1/2} \tag{42}$$

where

$$S = \int_{\theta_0}^{\theta_1} \phi \, d\theta \tag{43}$$

Furthermore, we denote by v_0 the absorption (or infiltration) rate, which is also the flow velocity at $x = 0$. Then

$$v_0 = di/dt = \tfrac{1}{2}St^{-1/2} \tag{44}$$

It is seen that the integral properties of the process [by which we mean functions such as $i(t)$, $v_0(t)$] are completely determined by the quantity, S, which we call the *sorptivity* of the porous medium or soil [84]. Evidently, S embodies in a single parameter the influence of capillarity on the transient flow processes that follow a step-function change in θ (or, more precisely, Ψ) at the surface of a porous mass. Strictly, we should write $S(\theta_0, \theta_1)$ or $S(\Psi_0, \Psi_1)$, since S has meaning only in relation to an initial state of the medium and an imposed boundary condition; however, we may usually omit the arguments of S without ambiguity. The dimensions of S are [length] $\times$ [time]$^{-1/2}$, and we use here the unit cm-sec$^{-1/2}$.

Clearly, $S(\theta_0, \theta_1)$ can be found by integration of the solution (36), so that the methods of Sect. IV.A enable $S(\theta_0, \theta_1)$ to be calculated for any porous medium or soil for which $D(\theta)$ is known in the θ range $\theta_0 \leqq \theta \leqq \theta_1$. Philip [85] discussed the variation of S with θ_0 with the aid of calculations of this type and of the "delta-function" model treated below (Sect. VII). There is no special difficulty in calculating S in cases (such as when water is supplied under positive pressure) where the analysis must be made, at least in part, with Ψ as the dependent variable. Philip [81] has studied the effect of the depth of the supplied (free) water on S, using the required extension to the method of Sect. IV.A; and he has related these results to the delta-function model [22].

2. *Intrinsic Sorptivity*

Under the assumption that absorption is the consequence of viscous flow induced by capillarity (which is certainly the primary mechanism of the absorption of water, unless it is supplied at very large [negative] Ψ values, or the soil is extremely colloidal in character), we may define the *intrinsic sorptivity*, $\mathscr{S}$, by the equation

$$\mathscr{S} = (\mu/\sigma)^{1/2} S \tag{45}$$

μ is the dynamic viscosity, and σ is the surface tension of the absorbate.

$\mathscr{S}$ is, then, like the intrinsic permeability κ, a characteristic of the internal geometry of the medium, and is independent of the properties of the absorbate. It is interesting to note that the dimensions of $\mathscr{S}$ are [length]$^{1/2}$, whereas those of κ are [length]^2. Different contact angles produce different geometrical relations of air-water interfaces, however, so that (for given θ_0 and θ_1) $\mathscr{S}$ depends on the contact angle in a very complicated way. The inclusion of the contact angle in the original definition of $\mathscr{S}$ by Philip [84] is not justified.

D. Illustrative Example

It is convenient for the purposes of exposition to introduce as an example the case of horizontal absorption into a particular soil, and to carry the illustration through the later developments below by presenting results for more complicated infiltration processes into the same soil. We choose for illustration the Yolo light clay of Moore [6]; we have already presented in Figs. 1–3 $K(\theta)$, $\Psi(\theta)$, and $D(\theta)$ functions for this soil. Figure 4 gives the $\phi(\theta)$

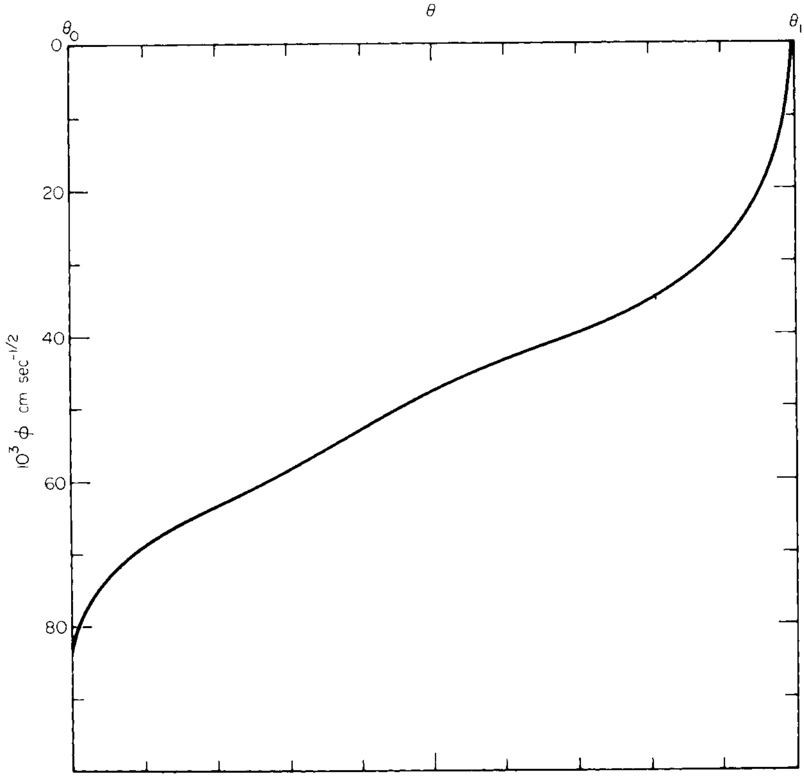

FIG. 4. Moisture profile during one-dimensional absorption in the dimensionless form $\phi(\theta)$, calculated for illustrative example of Yolo light clay of Figs. 1–3 with $\theta_0 = 0.2376$ and $\theta_1 = 0.4950$ [34].

function for the case where $\theta_0 = 0.2376$ and θ_1 is the "saturated" moisture content, 0.4950. For this system, the sorptivity, S, has the value 1.254×10^{-2} cm-sec$^{-1/2}$.

E. Two- and Three-Dimensional Series Solutions

Radially symmetrical m-dimensional absorption into a uniform soil with uniform initial moisture content is described by Eq. (46) subject to conditions (47):

$$\frac{\partial \theta}{\partial t} = r^{1-m} \frac{\partial}{\partial r}\left(D r^{m-1} \frac{\partial \theta}{\partial r}\right) \qquad \text{(46) [also Eq. (20)]}$$

$$\begin{aligned} t &= 0, \qquad \theta = \theta_0, \qquad r > r_0 \\ t &\geqq 0, \qquad \theta = \theta_1, \qquad r = r_0 \end{aligned} \tag{47}$$

$r = r_0$ is the surface of the cylindrical ($m = 2$) or spherical ($m = 3$) cavity from which water is supplied.

The following method of solution is due to Philip [86]. It is convenient to introduce the new variables

$$\rho = r/r_0, \qquad \tau = t/r_0^2 \tag{48}$$

which reduce Eqs. (46) and (47) to

$$\frac{\partial \theta}{\partial \tau} = \rho^{1-m} \frac{\partial}{\partial \rho}\left(D \rho^{m-1} \frac{\partial \theta}{\partial \rho}\right) \tag{49}$$

$$\begin{aligned} \tau &= 0, \qquad \theta = \theta_0, \qquad \rho > 1 \\ \tau &\geqq 0, \qquad \theta = \theta_1, \qquad \rho = 1 \end{aligned} \tag{50}$$

We seek solutions of Eq. (49) subject to conditions (50) of the form

$$\rho(\theta, \tau) = 1 + \phi_{*_1} \tau^{1/2} + \phi_{*_2} \tau + \phi_{*_3} \tau^{3/2} + \cdots \tag{51}$$

where each ϕ_* is a function of θ. *A priori* justification of Eq. (51) may be established along lines similar to those by which Philip [87] originally developed the series solution of the one-dimensional infiltration problem (Sect. V).

The identity

$$(\partial \theta / \partial \tau)_\rho \cdot (\partial \rho / \partial \theta)_\tau = -(\partial \rho / \partial \tau)_\theta \qquad (52)^1$$

converts Eq. (49) to a form with ρ as the dependent variable

$$-\frac{1}{m} \frac{\partial \rho^m}{\partial \tau} = \frac{\partial}{\partial \theta}\left(D \rho^{m-1} \frac{\partial \theta}{\partial \rho}\right) \tag{53}$$

[1] Here, and at a later point where it is also necessary for clarity, we use the notation $(\partial a/\partial b)_c$ to denote the partial derivative of variable a with respect to variable b with variable c held constant.

and an integration with respect to θ yields

$$-\frac{1}{m}\frac{\partial}{\partial \tau}\int_{\theta_0}^{\theta} \rho^m \, d\theta = D\rho^{m-1}\frac{\partial \theta}{\partial \rho} \tag{54}$$

Equation (54) depends on the result that $\lim_{\theta \to \theta_0} D\rho^{m-1}\, \partial\theta/\partial\rho = 0$ (physically that the flow velocity at infinity is zero for all finite t).

Substituting Eq. (51) in Eq. (54), and assuming that the expansion in $\tau^{1/2}$ of each side of Eq. (54) is convergent, we may equate coefficients of powers of $\tau^{1/2}$ and so obtain the following set of ordinary integrodifferential equations:

$$\int_{\theta_0}^{\theta} \phi_{*_1} \, d\theta = -2(D/\phi'_{*_1}) \tag{55}$$

$$\int_{\theta_0}^{\theta} \phi_{*_2} \, d\theta = [D\phi'_{*_2}/(\phi'_{*_1})^2] - (m-1)\int_{\theta_0}^{\theta} D \, d\theta \tag{56}$$

$$\int_{\theta_0}^{\theta} \phi_{*_3} \, d\theta = -\frac{2}{3}\left[\frac{D[(\phi'_{*_2})^2 - \phi'_{*_1}\phi'_{*_3}]}{(\phi'_{*_1})^3} - (m-1)\int_{\theta_0}^{\theta} D\phi_{*_1} \, d\theta\right] \tag{57}$$

and so forth. The primes in Eqs. (55)–(57) signify differentiation with respect to θ. In view of the second of conditions (50), the condition on the ϕ_*'s is

$$\theta = \theta_1, \qquad \phi_{*n} = 0, \qquad n = 1, 2, 3, 4, \ldots \tag{58}$$

Now, apart from a change of symbolism, Eqs. (55) and (58) are identical to Eqs. (37) and (38), so that we have at once that

$$\phi_{*_1}(\theta) \equiv \phi(\theta) \tag{59}$$

That is, the leading term of the series solution for m-dimensional absorption is exactly the solution for one-dimensional absorption. The physical explanation for this is simply that, when the depth of penetration of the m-dimensional absorption process is small compared to r_0, the process is essentially one-dimensional [88]. We may put the matter mathematically as follows: Rewriting Eq. (46) as

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial r}\left(D\frac{\partial \theta}{\partial r}\right) + \frac{(m-1)D}{r}\frac{\partial \theta}{\partial r} \tag{60}$$

we may show that, for t and $r - r_0$ small enough, the second term of the right side of Eq. (60) is negligibly small compared with the first. Its neglect, of course, reduces Eq. (60) to the one-dimensional form of Eq. (31).

It is evident, then, that [when we do not already know $\phi(\theta)$] Eq. (55) subject to conditions (58) may be solved by the methods of Sects. IV.A and IV.B.

Equations (56), (57), etc., may then be solved in turn. These are all linear equations and are readily amenable to rapid and accurate numerical solution.

The methods devised by Philip [87] to solve the rather similar set of linear equations which arise in the series solution of the one-dimensional infiltration equation may be used here also. We notice that the form of Eq. (56) is such that

$$\phi_{*2(m=3)} = 2\phi_{*2(m=2)} \tag{61}$$

with a consequent economy of labor of calculation if solutions are required for both $m = 2$ and $m = 3$.

The influence of r_0 on the dynamics of (at least the early stages of) the absorption process is implicit in Eq. (51), and is made explicit if we rewrite Eq. (51) in terms of r and t:

$$r(\theta, t) = r_0 + \phi_{*_1} t^{1/2} + \phi_{*_2}(t/r_0) + \phi_{*_3}(t^{3/2}/r_0{}^2) + \cdots \tag{62}$$

We may develop series expansions for the absorption rate and for the cumulative absorption in terms either of τ or t. It is convenient here to treat the matter in terms of absorption rate $v_0(t)$ and cumulative absorption $i(t)$, each based on unit area of the water-supply surface $r = r_0$. Then

$$i = \frac{\int_{r_0}^{\infty} (\theta - \theta_0) r^{m-1}\, dr}{r_0^{m-1}} = \frac{\int_{\theta_0}^{\theta_1} r^m \, d\theta}{m r_0^{m-1}} \tag{63}$$

Putting Eq. (62) in Eq. (63), expanding r^m, and integrating term by term, we obtain $i(t)$ as a power series in $t^{1/2}$. The leading terms are

$$i = S t^{1/2} + \frac{A_{*_2} t}{r_0} + \frac{A_{*_3} t^{3/2}}{r_0{}^2} + \cdots \tag{64}$$

S is, of course, the sorptivity defined in Sect. IV.C, and the A_{*_n} are appropriate integrals of the ϕ_{*_n}. Specifically,

$$A_{*_2} = \int_{\theta_0}^{\theta_1} \left[\frac{m-1}{2} \phi_{*_1}^2 + \phi_{*_2} \right] d\theta \tag{65}$$

and

$$A_{*_3} = \int_{\theta_0}^{\theta_1} \left[\frac{(m-1)(m-2)}{2.3} \phi_{*_1}^3 + (m-1)\phi_{*_1}\phi_{*_2} + \phi_{*_3} \right] d\theta \tag{66}$$

It follows from Eqs. (61) and (65) that

$$A_{*2(m=3)} = 2A_{*2(m=2)} \tag{67}$$

Differentiating Eq. (64) with respect to t yields the expansion for $v_0(t)$:

$$v_0 = \tfrac{1}{2} S t^{-1/2} + \frac{A_{*_2}}{r_0} + \frac{3A_{*_3} t^{1/2}}{2r_0{}^2} + \cdots \tag{68}$$

It must be understood that the time range of applicability of these expansions is definitely limited. We shall not offer here a detailed discussion of the question of their convergence. It seems sufficient to point out that the characteristic time[2] t_{geom}, based on S, $(\theta_1 - \theta_0)$, and the mean curvature (in three dimensions) of the supply surface, r_*^{-1},

$$t_{\text{geom}} = \left(\frac{r_*(\theta_1 - \theta_0)}{S}\right)^2 \tag{69}$$

gives the order of magnitude of t at which the m-dimensional geometry can be expected to swamp the initially one-dimensional character of the process. Note that r_* is $2r_0$ for $m = 2$ and is r_0 for $m = 3$. One expects *a priori* that the useful t range of these expansions will be rather less than t_{geom}; and, in fact, empirical experience suggests that this range is about $0.12t_{\text{geom}}$.

For the case $m = 3$, we may supplement this form of solution with the asymptotic (large-time) solution of Sect. IV.F.

F. Three-Dimensional Quasi-Analytical Large-Time Solution

It is readily shown that, in the limit as $t \to \infty$, the solution of Eq. (49) subject to conditions (50) for $m = 3$ is

$$\rho(\theta, \infty) = \int_{\theta_0}^{\theta_1} D\, d\theta \bigg/ \int_{\theta_0}^{\theta} D\, d\theta, \qquad \text{i.e.,} \quad r(\theta, \infty) = r_0 \int_{\theta_0}^{\theta_1} D\, d\theta \bigg/ \int_{\theta_0}^{\theta} D\, d\theta \tag{70}$$

It follows that, in the limit of t large, $v_0(t)$ approaches the steady value $v_0(\infty)$ given by Eq. (71):

$$v_0(\infty) = \int_{\theta_0}^{\theta_1} D\, d\theta / r_0 \tag{71}$$

This result [86] is a particular form of the more general results of Sect. IV.H. Its physical importance is that it exemplifies the existence of a large-time steady phase of three-dimensional absorption *even in the absence of gravity*; and it is useful as an illustration of the falsity of certain notions in the literature about the use of similarity solutions in m-dimensional systems (Sect. IV.I).

G. Illustrative Example: Two- and Three-Dimensional Absorption

Figures 5 and 6 present for our illustrative example the sequence of moisture profiles during two- and three-dimensional absorption as calculated from Eq. (51) (first three terms) and Eq. (70). To this accuracy of calculation, we have that, because of Eq. (61), each profile for $m = 2$ represents, for the

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[2] Discussion of physical processes in terms of characteristic time and length scales is a well-established procedure, although it has not been used much in the soil-water context [43, 89].

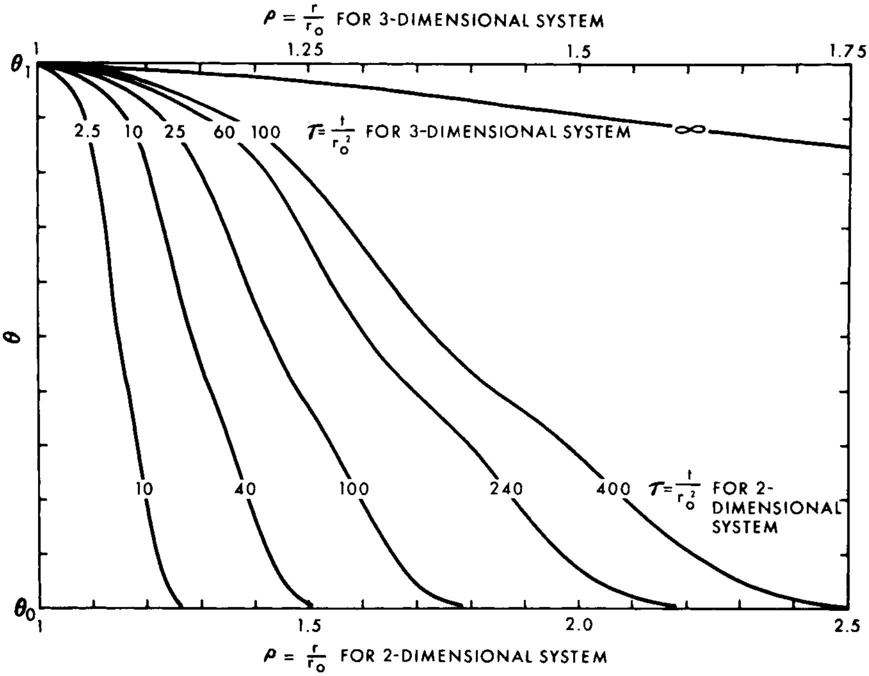


FIG. 5. Moisture profiles during two- and three-dimensional absorption, calculated for illustrative example of Yolo light clay of Figs. 1–3 with $\theta_0 = 0.2376$ and $\theta_1 = 0.4950$ [86]. Profiles computed from first three terms of Eq. (51), except for three-dimensional $\tau = \infty$ profile calculated from Eq. (71). Numerals on each profile represent time at which profile is realized in the form τ $(= t/r_0{}^2)$.

appropriately adjusted scale of ρ and value of τ, a profile for $m = 3$ also. The $m = 3$ profile for $\tau = \infty$, computed from Eq. (70), is shown in part in Fig. 5, and is compared with the $\tau = 100$ profile in Fig. 6. It will be seen that, whereas the corresponding one-dimensional absorption profiles preserve similarity, the profiles for $m = 2$ and 3 grow *relatively* flatter as τ increases. The flattening proceeds more rapidly for $m = 3$ than for $m = 2$.

H. Two- and Three-Dimensional Arbitrary Geometry: Quasi-Analytical Large-Time and Steady Solutions

Of the whole class of absorption processes described by Eq. (14), a large subclass has boundary conditions such that, in the limit as $t \to \infty$, the phenomenon approaches a steady state. For such steady states, Eq. (14) becomes

$$\nabla \cdot (D\, \nabla\theta) = 0 \tag{72}$$

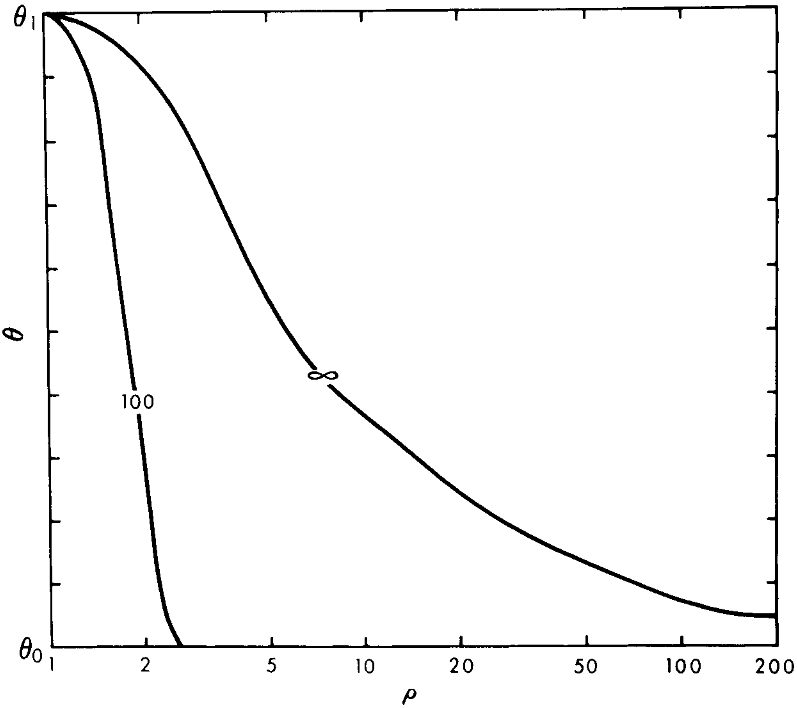


FIG. 6. As Fig. 5, showing only $\tau = 100$ and $\tau = \infty$ profiles for three-dimensional absorption [86].

The transformation [23, p. 11; 90–93]

$$\theta_* = \int_{\theta_0}^{\theta} D \, d\theta \tag{73}$$

reduces Eq. (72) to Laplace's equation

$$\nabla^2 \theta_* = 0 \tag{74}$$

For certain absorption phenomena (e.g., where water is under positive hydrostatic pressure in part of the flow region), the appropriate flow equation is that with Ψ, not θ, as the dependent variable [Eq. (13) with the second term of the right side omitted]. For steady states, this equation reduces to

$$\nabla \cdot (K \nabla \Psi) = 0 \tag{75}$$

and the transformation

$$\Psi_* = \int_{\Psi_0}^{\Psi} K \, d\Psi \tag{76}$$

again leads to Laplace's equation, now in the form

$$\nabla^2\Psi_* = 0 \tag{77}$$

The whole corpus of established analytical methods of solution of Laplace's equation in two and three dimensions is thus available for the establishment of quasi-analytical large-time (or steady) solutions of appropriate two- and three-dimensional absorption problems.

It remains for us to specify these problems. It is necessary that their boundary conditions have definite limits as $t \to \infty$, and that these "limiting" conditions be such that a solution of Laplace's equation exists for them which is neither singular nor trivial.

Turning, in particular, to absorption in systems that are infinite in two or three dimensions, *but in which the supply surface is finite*, we remark that it can be shown that the steady state always exists in three dimensions, but never in two. This result holds whether the boundary condition at the supply surface is of the concentration type (as for most of the processes considered in this paper) or of the flux type.

It is not our purpose here to make an exhaustive exploration of these solutions, but we indicate two examples. As we have noted already, the solution of Sect. IV.F is of this class. A second example is that of absorption into a three-dimensional semi-infinite region from a plane circular region of radius r_0 at the surface of which the moisture content is maintained at the value θ_1. The required large-time solution follows at once from the relevant solution of Laplace's equation [23, p. 215]. It suffices here to remark that, with v_0 based on unit area of the supply surface,

$$v_0(\infty) = (4/\pi r_0) \int_{\theta_0}^{\theta_1} D\, d\theta \tag{78}$$

The large-time *total* absorption rate is then

$$q(\infty) = 4r_0 \int_{\theta_0}^{\theta_1} D\, d\theta \tag{79}$$

These results should be compared with those for absorption from a *hemispherical* supply surface of radius r_0 into a three-dimensional semi-infinite region (which follow at once from Sect. IV.F). In that case,

$$v_0(\infty) = (r_0)^{-1} \cdot \int_{\theta_0}^{\theta_1} D\, d\theta, \qquad q(\infty) = 2\pi r_0 \int_{\theta_0}^{\theta_1} D\, d\theta \tag{80}$$

I. Similarity Solutions in Absorption Problems

Similarity methods form an important avenue of mathematical attack on nonlinear problems; as we have seen, the one-dimensional absorption solution is a similarity solution. These methods are better known in more conventional branches of fluid mechanics than they seem to be in soil-water studies, and several confusions in the mathematical study of absorption have arisen from the misuse of similarity methods.

Miller and Miller [65] correctly observed that the cases $m = 2, 3$, $r_0 = 0$ lead to a similarity reduction of Eq. (46) subject to conditions (47), but that, for both values of m, a singularity at $r = 0$ implies a trivial "no-flow" solution in the physically meaningful situation where θ_1 is required to be finite. They went on, however, to suggest that experiments with an impressed flow rate proportional to $t^{1/2}$ and a small but nonzero r_0 would yield similarity. [It appears that they, in fact, intended to propose a constant (t^0) flow rate for $m = 2$ and a flow as $t^{1/2}$ for $m = 3$.] Their proposal depends on the heuristic idea that, if something is true for r_0 zero, it will be more or less true for r_0 "small": in fact, the singular and degenerate nature of the solution for $r_0 = 0$ and θ_1 finite completely invalidates the argument. It will be clear from the relevant preceding solutions (and from later supplementary results) that absorption solutions for $m = 2, 3$ with r_0 nonzero are never, even approximately, of the form

$$r(\theta, t) = f(\theta)t^{1/2}$$

as suggested by Miller and Miller [65]. For $m = 2$, in fact, v_0 varies initially as $t^{-1/2}$ and finally as $(\log t)^{-1}$; and, for $m = 3$, v_0 varies initially as $t^{-1/2}$ and finally as t^0: compare the "similarity predictions" that the behavior is as t^0 for $m = 2$ and as $t^{1/2}$ for $m = 3$.

The misuse of similarity methods seems to arise often from neglect of the following basic principle [86, 94]. A similarity substitution usefully reduces the number of independent variables in a partial differential equation only when the variables removed from the equation are removed also from all the governing conditions by the same substitution. Singh [95] and Singh and Franzini [96] attempted to study absorption for $m = 2$ by the use of similarity methods, but the disregard of this principle vitiates the results [86, 97]. Swartzendruber [98] and Philip [91] pointed out that the additive-variables substitution used by Kobayashi [99, 100] fails for essentially the same reason.

It appears that the *Boltzmann transformation* (33) is the one useful similarity substitution in studies of absorption with concentration-type boundary conditions. Kobayashi [100] claimed to have developed a diversity of substitutions, but they include nothing new that is both correct and useful [91]. The "three-dimensional" substitutions are simply the one-dimensional Boltzmann transformation in an inappropriate coordinate system, and other substitutions claimed as new are simply members of the indefinitely large class of "pseudotransformations" of the form $f(\phi)$ (with f a monotonic function of ϕ) which are totally equivalent to the Boltzmann transformation.

We see later in Sect. IX that other forms of similarity substitution are useful in connection with absorption processes involving a flux-type boundary condition.

V. Exact Solutions of Infiltration Equations

A. One-Dimensional Series Solution

Infiltration into an effectively semi-infinite, homogeneous, one-dimensional system of uniform initial moisture content is described by Eq. (81) subject to conditions (82):

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z}\left(D \frac{\partial \theta}{\partial z}\right) - \frac{dK}{d\theta} \cdot \frac{\partial \theta}{\partial z} \qquad \text{(81) [also Eq. (22)]}$$

$$\begin{aligned} t &= 0, & z &> 0, & \theta &= \theta_0 \\ t &\geqq 0, & z &= 0, & \theta &= \theta_1 \end{aligned} \qquad (82)$$

Philip [33, 34, 87] developed the following method of solution of Eq. (81) subject to conditions (82). The solution is sought in the form

$$z(\theta, t) = \phi_1 t^{1/2} + \phi_2 t + \phi_3 t^{3/2} + \phi_4 t^2 + \cdots \qquad (83)$$

where each of the ϕ's is a function of θ. The form of Eq. (83) was found by Philip [87] by a lengthy process of successive approximation. The reader who seeks a full *a priori* justification is referred to that paper. Equation (83) is, in fact, the immediate outcome of considering the solution of Eq. (81) subject to conditions (82) as a perturbation of the solution of Eq. (31) subject to conditions (32) (with z substituted for x). It will be noted that, here, as in Sect. IV.E, the method involves taking the space coordinate as a dependent variable, with θ as an independent variable.

Accordingly, we use the identity

$$(\partial\theta/\partial t)_z \cdot (\partial z/\partial \theta)_t = -(\partial z/\partial t)_\theta \qquad (84)$$

to reduce Eq. (81) to the form

$$-\frac{\partial z}{\partial t} = \frac{\partial}{\partial \theta}\left(D \frac{\partial \theta}{\partial z}\right) - \frac{dK}{d\theta} \qquad (85)$$

Integrating with respect to θ, we obtain

$$-\frac{\partial}{\partial t}\int_{\theta_0}^{\theta} z\, d\theta = D\frac{\partial \theta}{\partial z} - (K - K_0) \qquad (86)$$

We write K_0 for $K(\theta_0)$. Equation (86) uses the result $\lim_{\theta\to\theta_0} D\, \partial\theta/\partial z = 0$ for all t, which is implicit in the first of conditions (82).

Putting Eq. (83) in Eq. (86), and assuming that the expansion in $t^{1/2}$ of each side of Eq. (86) is convergent, we may equate coefficients of powers of $t^{1/2}$ and so obtain the following set of ordinary integrodifferential equations:

$$\int_{\theta_0}^{\theta} \phi_1 \, d\theta = -2 \frac{D}{\phi_1'} \tag{87}$$

$$\int_{\theta_0}^{\theta} \phi_2 \, d\theta = \frac{D\phi_2'}{(\phi_1')^2} + (K - K_0) \tag{88}$$

$$\int_{\theta_0}^{\theta} \phi_3 \, d\theta = \frac{2D}{3} \left[\frac{\phi_3'}{(\phi_1')^2} - \frac{(\phi_2')^2}{(\phi_1')^3} \right] \tag{89}$$

$$\int_{\theta_0}^{\theta} \phi_4 \, d\theta = \frac{D}{2} \left[\frac{\phi_4'}{(\phi_1')^2} - \frac{(\phi_2')^2}{(\phi_1')^3} \left\{ 2 \frac{\phi_3'}{\phi_2'} - \frac{\phi_2'}{\phi_1'} \right\} \right] \tag{90}$$

and so on, the equation for $\phi_n (n \geqq 3)$ being of the form

$$\int_{\theta_0}^{\theta} \phi_n \, d\theta = \frac{2D}{n} \left[\frac{\phi_n'}{(\phi_1')^2} - R_n(\theta) \right] \tag{91}$$

where R_n may be determined from $\phi_1, \ldots, \phi_{n-1}$. It will be understood that the prime in Eqs. (87)–(91) signifies differentiation with respect to θ. In view of the second of conditions (82), the condition on the ϕ's is

$$\theta = \theta_1, \qquad \phi_n = 0, \qquad n = 1, 2, 3, 4, \ldots \tag{92}$$

Now, apart from a difference of symbolism, Eqs. (87) and (92) are identical with Eqs. (37) and (38), so that we have at once that

$$\phi_1(\theta) \equiv \phi(\theta) \tag{93}$$

Evidently, if we do not already know $\phi(\theta)$, we may solve Eq. (87) subject to conditions (92) by the methods of Sects. IV.A and IV.B.

We see that the leading term of the series solution for one-dimensional infiltration is exactly the solution for one-dimensional absorption. In physical terms, the explanation is that, for small t, the potential gradients in the flow region due to capillarity are much greater than unity, the value of the potential gradient due to gravity. In mathematical terms, we may show that, for t and z small enough, the second term of the right side of Eq. (81) is negligibly small compared with the first; and the neglect of this term reduces Eq. (81) to Eq. (31).

With ϕ_1 known, we may solve Eq. (88) subject to conditions (92) for ϕ_2; then, with ϕ_2 known, we may solve Eq. (89) subject to conditions (92) for ϕ_3; and so on. The various equations (88)–(91) are linear, and may be rapidly and accurately solved by the numerical method of Philip [87]. In brief, the method uses appropriate finite difference forms and forward integration initiated by an assumed value of $\int_{\theta_0}^{\theta_1} \phi_n \, d\theta$. A recent account of this work [101], missing the point that these equations are linear, implies that the method of solution is iterative; however, because the equations are linear, iteration is not needed. One simply performs the calculation for two arbitrary values of $\int_{\theta_0}^{\theta_1} \phi_n \, d\theta$, and linear interpolation then yields the "exact" solution. The precision of the method is improved by use of the Richardson h^2-extrapolation [80] and by the use of analytical methods in the neighborhood of the "infinite tail," which, in general, occurs as $\theta \to \theta_0$.

The total increase of moisture content in the semi-infinite region during the time interval 0 to t equals the difference between the time integrals of the flow velocity at $x = 0$ and that at infinity. That is,

$$\int_0^\infty (\theta - \theta_0)\, dx = \int_{\theta_0}^{\theta_1} x \, d\theta = i(t) - K_0 t \tag{94}$$

where $i(t)$ is the cumulative infiltration. Putting Eq. (83) in Eq. (94), we obtain the series for $i(t)$ [34]:

$$i(t) = St^{1/2} + (A_2 + K_0)t + A_3 t^{3/2} + A_4 t + \cdots \tag{95}$$

Here S is the sorptivity, and

$$A_n = \int_{\theta_0}^{\theta_1} \phi_n \, d\theta \tag{96}$$

Differentiating Eq. (95) with respect to t, we obtain the series for the infiltration rate, $v_0(t)$ [34]:

$$v_0(t) = \tfrac{1}{2}St^{-1/2} + (A_2 + K_0) + \tfrac{3}{2}A_3 t^{1/2} + 2A_4 t + \cdots \tag{97}$$

A rigorous treatment of the convergence of these series seems rather difficult, and we limit ourselves here to some general remarks. In this connection, it is useful to introduce a characteristic time of the infiltration process, t_{grav}, based on S and $K_1 - K_0$ [we write K_1 for $K(\theta_1)$],

$$t_{\text{grav}} = \left(\frac{S}{K_1 - K_0}\right)^2 \tag{98}$$

The order of magnitude of the t value at which the effect of gravity on the process can be expected to be as great as that of capillarity is given by t_{grav}; and one expects *a priori* that the t range of useful convergence for expansions of Eqs. (83), (95), and (97) will be of about this magnitude.

It is observed empirically in numerical examples that

$$n > 2, \qquad A_n/S > (A_2/S)^{n-1} \tag{99}$$

so that series (95) may be shown, by comparison with the geometric series, to converge for

$$t < (S/A_2)^2 \tag{100}$$

Now, for the usual case with $K_0/K_1 \ll 1$, it is found both empirically and by analysis (cf. Sects. VI and VIII) that $A_2 \approx \frac{1}{2}(K_1 - K_0)$. Inequality (100) then becomes

$$t < 4t_{\text{grav}} \tag{101}$$

One expects the practical limit of convergence to be given by inequality (101) with 4 replaced by a smaller numerical coefficient. Empirical experience suggests, in fact, that a suitable criterion for *practical convergence* is

$$t \leqq t_{\text{grav}} \tag{102}$$

This method applies in a slightly modified form [81] to the case of water supplied under positive hydrostatic pressure. Equation (81) is replaced by the relevant form of Eq. (13), at least in the region of positive Ψ.

B. One-Dimensional Asymptotic (Large-Time) Solution

We supplement the preceding solution of Eq. (81) subject to conditions (82) with the asymptotic solution valid for large t obtained by Philip [73]. We seek a solution of the form

$$z(\theta, t) = u(t - t_0) + \zeta(\theta) \tag{103}$$

Substituting Eq. (103) in Eq. (86), we obtain

$$u(\theta - \theta_0) = (K - K_0) - D(d\theta/d\zeta) \tag{104}$$

For t sufficiently large, we evaluate u in Eq. (104) with the aid of the result

$$\lim_{t\to\infty} \lim_{\theta\to\theta_1} \partial\theta/\partial z \equiv 0 \tag{105}$$

which is the consequence of theorems established by Philip [73]. Equation (105) implies that, if Eq. (103) holds for indefinitely large t, $\lim_{\theta\to\theta_1} d\theta/d\zeta = 0$. It follows that

$$u = (K_1 - K_0)/(\theta_1 - \theta_0) \tag{106}$$

Putting Eq. (106) in Eq. (104) and integrating, we obtain

$$\zeta(\theta) = (\theta_1 - \theta_0) \int_\theta^{\theta_a} \frac{D\, d\theta}{(K_1 - K_0)(\theta - \theta_0) - (K - K_0)(\theta_1 - \theta_0)} \tag{107}$$

where the zero of ζ is taken at $\theta = \theta_a$. Usually, θ_a is conveniently taken as $\theta_1 - \Delta\theta$, where $\Delta\theta$ is a definite small positive quantity. It is usually necessary to take the zero of ζ at θ values other than θ_1 and θ_0, because, in general, ζ has singularities for both these θ values [73]. Except in the special case where $D \to 0$ sufficiently rapidly as $\theta \to \theta_0$, which is treated by Philip [48], we have that

$$\lim_{\theta\to\theta_1} \zeta = -\infty, \qquad \lim_{\theta\to\theta_0} \zeta = +\infty \tag{108}$$

The singular behavior of ζ is not recognized in several accounts of the integration of Eq. (107) [101–103]. Evidently, $\zeta(\theta)$ for $\theta_0 < \theta < \theta_1$ can be evaluated simply by numerical integration; Philip [73] has shown how the accuracy of the calculation can be improved near $\theta = \theta_0$ and θ_1 by the use of analytical methods.

Because $d^2K/d\theta^2$ is inherently nonnegative and is positive for (at least) almost all θ, the denominator of the integrand of Eq. (107) is positive for $\theta_0 < \theta < \theta_1$. It is of some interest that the phenomenon of the "profile at infinity"—i.e., that, asymptotically, the moisture profile becomes of the constant shape $\zeta(\theta)$, moving downward with constant velocity, u—is inherent in the nonlinearity of Eq. (81). For the corresponding linearized equation (with K a linear function of θ), the profile at infinity does not occur, the gradients of θ simply becoming progressively smaller everywhere in the wetted region as t increases (Sect. VI.C). Equation (107) applies also to the case of water supplied under positive hydrostatic pressure [81].

The similar and independent analysis by Youngs [102] takes Eq. (105) for granted and erroneously puts $u = K_1/\theta_1$. Irmay [104] recognized that the form of Eq. (103) would constitute an asymptotic large-time solution of Eq. (81). [He had derived an equation formally equivalent to Eq. (81) from a physically unacceptable model of the flow process.] Childs [103] gave an elegant restatement of the theorems on moisture profile development which underlay the original analysis [73]. His improved treatment of the possible influence of capillary hysteresis makes use of the independent domain theory [66], discussed in Sect. II.D.6.

It follows from the preceding analysis [and from Eq. (105), in particular] that, as t increases indefinitely, the infiltration rate decreases monotonically to its final asymptotic value, $v_0(\infty)$,

$$v_0(\infty) = K_1 \tag{109}$$

These various results provide means of estimating $z(\theta, t)$ and $i(t)$ for large t values. Fairly simple interpolation and matching techniques to connect the series and asymptotic solutions appear to be adequate in practice. The reader is referred to Philip [73] for details. Farrell [105] has suggested some refinements.

C. Illustrative Example: One-Dimensional Infiltration

Figure 7 presents for our illustrative example the functions $\phi_2(\theta)$, $\phi_3(\theta)$, $\phi_4(\theta)$. It will be understood that $\phi(\theta)$ of Fig. 4 is the relevant $\phi_1(\theta)$ function. The calculated values of A_2, A_3, A_4 are, respectively, 4.654×10^{-6} cm-sec^{-1}, 1.405×10^{-9} cm-sec$^{-3/2}$, and 8.961×10^{-14} cm-sec^{-2}. $(K_1 - K_0) = 12.288 \times 10^{-6}$ cm-sec^{-1}. $t_{\text{grav}} = 1.041 \times 10^6$ sec. Figure 8 gives the profile at infinity,

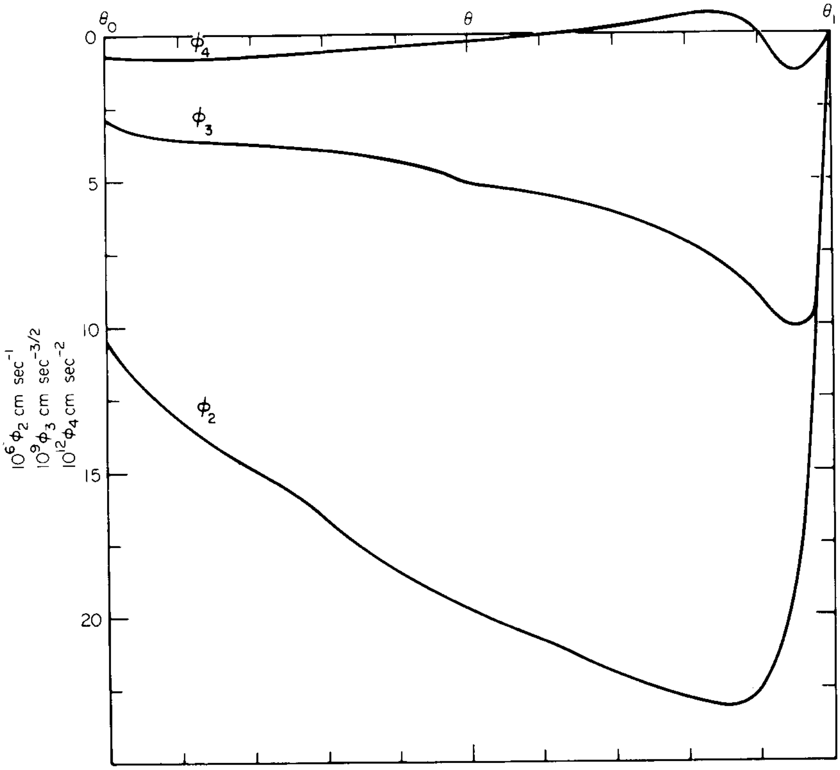


FIG. 7. The functions $\phi_2(\theta)$, $\phi_3(\theta)$, $\phi_4(\theta)$, calculated for one-dimensional infiltration in Yolo light clay of Figs. 1–3 with $\theta_0 = 0.2376$ and $\theta_1 = 0.4950$ [34].

$\zeta(\theta)$, for our example; θ_a is taken as $(0.95\theta_1 + 0.05\theta_0)$. $u = 47.74 \times 10^{-6}$ cm-sec^{-1}. Figure 9 presents the sequence of moisture profiles during one-dimensional infiltration. The profiles for $t \leq 10^6$ sec are calculated from the first four terms of series (83); those for larger values of t are based on asymptotic solution (103).

D. Two- and Three-Dimensional Small-Time Solutions

The techniques of Sects. V.A and V.B cannot be extended satisfactorily to infiltration in two and three dimensions, which involves at least two space coordinates.

For the early stages of infiltration into such systems (with a concentration boundary condition at the absorbing surface), limited use can be made, however, of the absorption solutions of Sect. IV. We discuss the matter in terms of

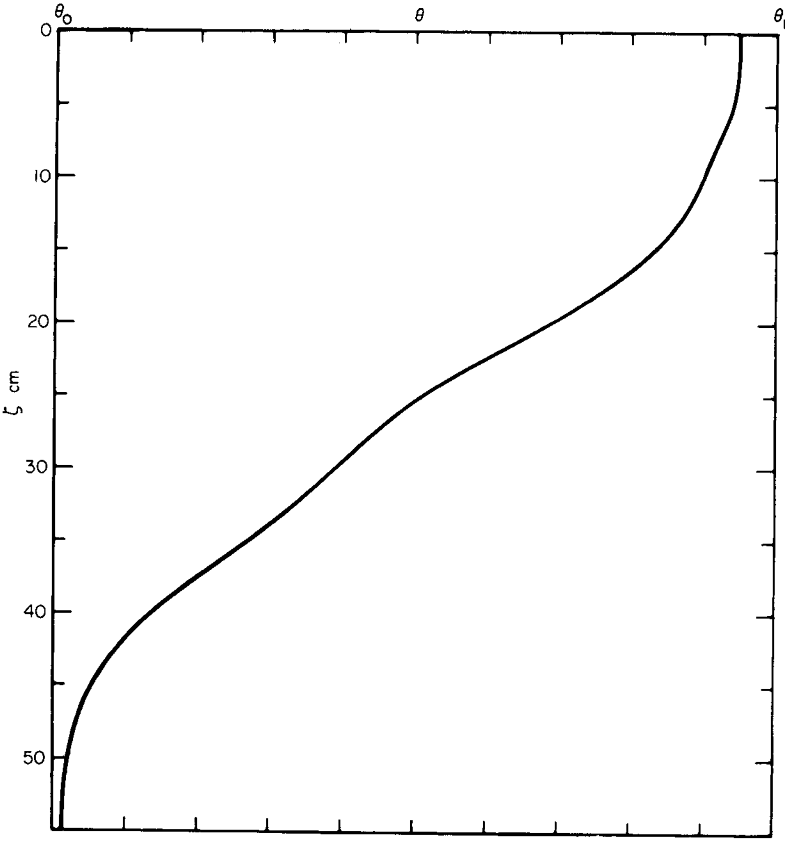


FIG. 8. The "profile at infinity," $\zeta(\theta)$, for illustrative example of one-dimensional infiltration [73]. $\theta_a = 0.95\theta_1 + 0.05\theta_0$.

the characteristic times t_{geom} [Eq. (69)] and t_{grav} [Eq. (98)]. When the corresponding absorption solution is known, it will provide a useful estimate of behavior for $t \leqq ct_{\text{grav}}$, where c is perhaps about 0.01. For $t_{\text{geom}} \ll t_{\text{grav}}$, however, absorption solutions found as expansions in $t^{1/2}$ may cease to be useful at t values much smaller than ct_{grav}. On the other hand, for $t_{\text{grav}} \ll t_{\text{geom}}$, ct_{grav} will be so small that the one-dimensional absorption solution will be virtually indistinguishable from the absorption solution for the actual geometry. The one-dimensional solution is, in this limited sense, useful for infiltration in geometries for which no exact absorption solution is known. It should be noted that, when the supply surface is other than cylindrical or spherical, some care must be given to the choice of the r_*^{-1} used to define t_{geom}. The maximum mean curvature of the supply surface will often be the appropriate value.

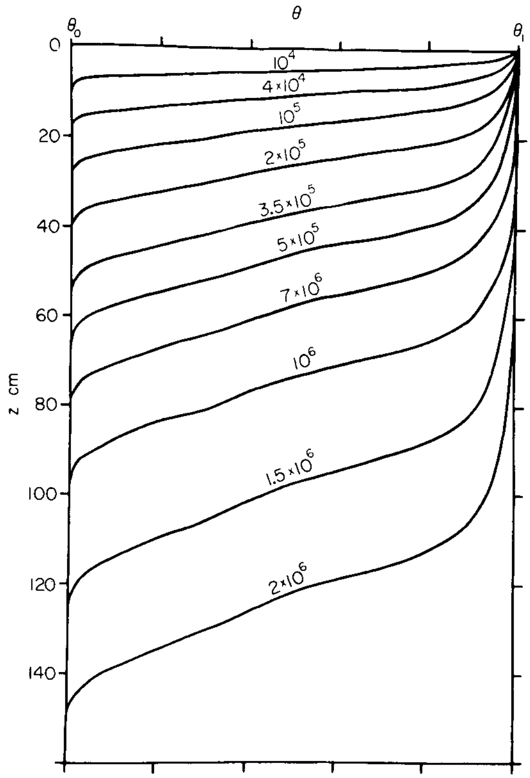


FIG. 9. Computed moisture profiles for illustrative example of one-dimensional infiltration. Numerals on each profile represent value of t (sec) at which profile is realized. Profiles for $t \leqq 10^6$ calculated from first four terms of series (83); those for larger t are based on Eq. (103) [34, 73].

E. TWO- AND THREE-DIMENSIONAL ARBITRARY GEOMETRY: QUASI-ANALYTICAL LARGE-TIME AND STEADY SOLUTIONS

In this subsection, we discuss the subclass of infiltration processes described by Eq. (21) for which, in the limit as $t \to \infty$, the phenomenon approaches a steady state. It is significantly wider than the subclass of absorption processes with the same property; thus, infiltration from a finite supply surface into a region infinite in two dimensions asymptotically approaches a steady state at large t, whereas the corresponding absorption problems do not have steady solutions.

For steady states, Eq. (21) becomes

$$\nabla \cdot (D\,\nabla\theta) = \frac{dK}{d\theta} \cdot \frac{\partial \theta}{\partial z} \tag{110}$$

and transformation (73) reduces this to

$$\nabla^2 \theta_* = \frac{1}{D} \frac{dK}{d\theta} \cdot \frac{\partial \theta_*}{\partial z} \tag{111}$$

Now, if

$$\frac{1}{D} \frac{dK}{d\theta} = \text{const} = \alpha \tag{112}$$

important simplifications follow, so that it is of interest to examine the physical implications of Eq. (112). We find by combining Eqs. (11) and (112) that Eq. (112) is true when

$$K \propto e^{\alpha \Psi} \tag{113}$$

Even though it cannot be claimed that Eq. (113) is universally exact, it does model in a reasonably convincing way the quite generally observed rapid and nonlinear decrease of K with Ψ. For many soils, Eq. (113) does, in fact, represent $K(\Psi)$ fairly well over Ψ ranges of interest in soil-water studies. α has the dimensions $[\text{length}]^{-1}$ and is conveniently expressed in cm^{-1}. Typically, α is about 0.01 cm^{-1}, and the range of values 0.05 to 0.002 cm^{-1} seems likely to cover most applications. α is a measure of the relative importance of gravity and capillarity for soil-water movement in the particular soil. Fine-textured soils, where capillarity tends to predominate, have small α values; and coarse-textured soils, where gravity effects manifest themselves most readily, have large α values.

Putting Eq. (112) in Eq. (111), we obtain the linear equation

$$\nabla^2 \theta_* = \alpha(\partial \theta_* / \partial z) \tag{114}$$

It will be seen that transformation (73), supplemented by Eq. (112), enables us to reduce the highly nonlinear Eq. (110) to a linear form *while still retaining the basically nonlinear character of the flow process*. Evidently, the exploration of solutions of Eq. (114) can be expected to provide considerable illumination of the problems of steady infiltration in two and three dimensions.

Problems of this type (including seepage from trenches, canals, basins, etc.) have been treated classically as problems of saturated flow with a free surface (e.g., Muskat [106], Polubarinova-Kochina [107]). That mode of analysis may be relevant where a water table exists at shallow depth; but it is definitely inappropriate in situations, common in arid and semi-arid environments, where there is no shallow water table. In such cases, we seek to study the steady regime set up by prolonged infiltration (or seepage) into *an initially unsaturated system*. The phenomenon is essentially one of flow in unsaturated soil, and a formalism based on Eq. (114) promises to be both physically apposite and mathematically amenable. It is of some interest that the unsaturated analysis, which might be expected *a priori* to be more complicated than the classical analysis, is the simpler in the sense that it is free of the difficulties connected with the free surface in the classical approach.

The pertinence of Eq. (114) has been recognized only recently [91–93], and the exploration of relevant solutions has not yet been taken very far. Here we limit specific discussion to two basic solutions, the source solutions in two and three dimensions, with θ_0 so small that K_0 is negligibly small.

1. Source Solution in Two Dimensions

The two-dimensional form of Eq. (114) is

$$\frac{\partial^2\theta_*}{\partial x^2} + \frac{\partial^2\theta_*}{\partial z^2} = \alpha\frac{\partial\theta_*}{\partial z} \tag{115}$$

It follows from Eq. (73) that, for infiltration into a two-dimensionally infinite region with initial moisture content θ_0, condition (116) must be satisfied:

$$\lim_{(x^2+z^2)^{1/2}\to\infty} \theta_* = 0 \tag{116}$$

The solution of Eq. (115) subject to condition (116) for the case of a continuous source of strength Q_2 cm^2-sec^{-1} at $x = 0$, $z = 0$ is well known [23, p. 267]. It is

$$\theta_* = (Q_2/2\pi)e^{\alpha z/2}\mathbf{K}_0[\alpha(x^2 + z^2)^{1/2}/2] \tag{117}$$

$\mathbf{K}_0$ is the modified Bessel function of the second kind of zero order. We may rewrite Eq. (117) as

$$\vartheta_2 = e^Z\mathbf{K}_0[(X^2 + Z^2)^{1/2}] \tag{118}$$

where

$$\vartheta_2 = 2\pi\theta_*/Q_2, \qquad X = \alpha\,|x|/2, \qquad Z = \alpha z/2 \tag{119}$$

This solution is presented in the form $\vartheta_2(X, Z)$ in Fig. 10.

This basic result can be used, as it stands, to develop solutions for infiltration from the surface of approximately circular cylindrical cavities in infinite regions; and appropriate source-sink distributions based on it yield solutions for infiltration from the surface of furrows of various shapes into semi-infinite regions bounded by an approximately horizontal upper surface.

2. Source Solution in Three Dimensions

The three-dimensional axisymmetric form of Eq. (114) is

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\theta_*}{\partial r}\right) + \frac{\partial^2\theta_*}{\partial z^2} = \alpha\frac{\partial\theta_*}{\partial z} \tag{120}$$

r is here $(x^2 + y^2)^{1/2}$. For axisymmetric infiltration into a three-dimensionally infinite region with initial moisture content θ_0, the final steady solution must satisfy the condition (121):

$$\lim_{(r^2+z^2)^{1/2} \to \infty} \theta_* = 0 \tag{121}$$

The solution of Eq. (120) subject to condition (121) for the case of a continuous source of strength Q_3 cm^{3}sec^{-1} at $r = 0$, $z = 0$ is [23, p. 267]

$$\theta_* = \frac{Q_3}{4\pi(r^2 + z^2)^{1/2}} \exp\left\{-\frac{\alpha}{2}[(r^2 + z^2)^{1/2} - z]\right\} \tag{122}$$

We may rewrite this as

$$\vartheta_3 = \frac{\exp[Z - (R^2 + Z^2)^{1/2}]}{(R^2 + Z^2)^{1/2}} \tag{123}$$

where

$$\vartheta_3 = 8\pi\theta_*/\alpha Q_3, \qquad R = \alpha r/2 \tag{124}$$

Figure 11 gives this solution in the form $\vartheta_3(R, Z)$.

Like Eq. (117), Eq. (122) leads to a diversity of solutions of interest. As it stands, it yields solutions for infiltration from the surface of approximately spherical cavities in infinite regions; and source-sink distributions based on it yield solutions for infiltration from the surface of basins of various shapes into semi-infinite regions bounded by an approximately horizontal upper surface.

3. *Other Solutions of Eq. (114)*

There remains much scope for the study of Eq. (114) by various methods. A boundary-layer technique neglecting $\partial^2\theta_*/\partial z^2$ {or, for the untransformed equation, $(\partial/\partial z)[D(\partial\theta/\partial z)]$}, similar to those employed in other contexts (e.g., Philip [94], Sutton [108], Wooding [109, 110]), is feasible and promises to give accurate results for the cases of infiltration from the surfaces of deep vertical slits and boreholes.

VI. Linearized Solutions

Apart from a limited range available for other classes of problem (see Sect. IX), we have exhausted the known solutions that may be applied to soils of arbitrary hydrologic characteristics. We go on to consider two classes of analytical solutions which are of some interest in their own right, and which are the basis of a comparison method for estimating integral properties of soils with arbitrary characteristics. We first discuss linearized solutions [74, 86, 111].

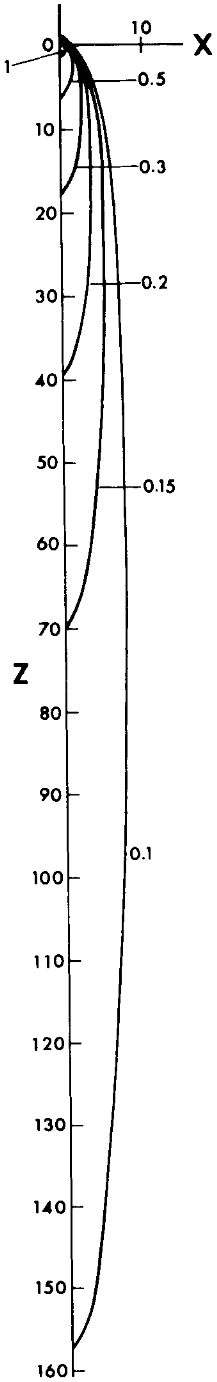


FIG. 10.

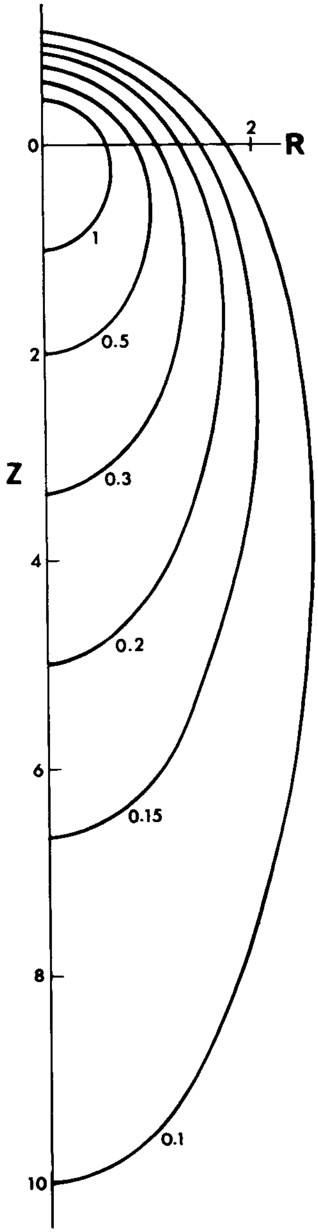


FIG. 11.

FIG. 10. Source solution for two-dimensional infiltration as $t \to \infty$, in the dimensionless form $\vartheta_2(X, Z)$. Numerals on the curves are values of ϑ_2.

FIG. 11. Source solution for three-dimensional infiltration as $t \to \infty$, in the dimensionless form $\vartheta_3(R, Z)$. Numerals on the curves are values of ϑ_3.

A. Principles of Method

The linearized form of Eq. (21) is

$$\partial\vartheta/\partial t = D_* \nabla^2\vartheta - k(\partial\vartheta/\partial z) \tag{125}$$

a linear Fokker–Planck equation. The equation is exact, and $\vartheta \equiv \theta$, for the case where

$$D(\theta) = \text{const} = D_* \tag{126}$$

and

$$dK/d\theta = \text{const} = k \tag{127}$$

As we have emphasized, D and $dK/d\theta$ are, for real media and soils, strongly varying functions of θ, so that solutions of Eq. (125) do not, in general, give an accurate detailed description of the phenomena that concern us here. We shall show, however, that Eq. (125) yields useful estimates of *integral properties* of these phenomena, $v_0(t)$ and $i(t)$. By choosing D_* so as to match the known integral behavior of absorption (and infiltration) processes in the limit as $t \to 0$, and by choosing k to match infiltration behavior in the limit as $t \to \infty$, we, essentially, use Eq. (125) as the basis of extrapolation and interpolation procedures that take known exact nonlinear solutions as reference points. In certain cases, which we shall identify, there exists a definite one-to-one relationship between θ and ϑ; and, in general, there is at least a loose ordinal relation between these two quantities.

B. Absorption Solutions

1. One-Dimensional Absorption

The linearized form of Eq. (31) subject to conditions (32) is

$$\frac{\partial\vartheta}{\partial t} = D_* \frac{\partial^2\vartheta}{\partial x^2} \tag{128}$$

subject to the conditions

$$\begin{aligned} t &= 0, \qquad x > 0, \qquad \vartheta = \theta_0 \\ t &\geqq 0, \qquad x = 0, \qquad \vartheta = \theta_1 \end{aligned} \tag{129}$$

The solution is

$$\vartheta = \theta_0 + (\theta_1 - \theta_0)\,\mathrm{erfc}[\phi/2D_*^{1/2}] \tag{130}$$

and

$$i = 2(D_*/\pi)^{1/2}(\theta_1 - \theta_0)t^{1/2} \tag{131}$$

Matching Eq. (131) with Eq. (42), we have

$$D_* = \pi S^2/4(\theta_1 - \theta_0)^2 \tag{132}$$

The matching equation (132) is valid for all absorption and infiltration processes induced by a step-function change in θ (or Ψ) at the supply surface (since all such processes are indistinguishable from one-dimensional absorption in the limit as $t \to 0$).

Putting Eq. (132) in Eq. (130), we have

$$\vartheta(\theta) = \theta_0 + (\theta_1 - \theta_0)\operatorname{erfc}\frac{(\theta_1 - \theta_0)\phi(\theta)}{\pi^{1/2}S} \tag{133}$$

where $\phi(\theta)$ is the solution of Eq. (34) subject to conditions (35). Equation (133) defines the transformation between ϑ and θ which makes Eq. (130) *exact* for a medium with arbitrary $D(\theta)$. Applied solely to this problem, the linearization technique is exact, but yields no new information. It is useful only when we now apply it to more complicated problems.

2. Two- and Three-Dimensional Absorption

Linearizing Eqs. (46) and (47) gives

$$\frac{\partial\vartheta}{\partial t} = D_* r^{1-m}\frac{\partial}{\partial r}\left(r^{m-1}\frac{\partial\vartheta}{\partial r}\right) \tag{134}$$

$$\begin{aligned} t &= 0, \qquad \vartheta = \theta_0, \qquad r > r_0 \\ r &\geqq 0, \qquad \vartheta = \theta_1, \qquad r = r_0 \end{aligned} \tag{135}$$

In the discussion that follows, we use the dimensionless variables

$$\begin{aligned} V &= \frac{v_0 r_0}{D_*(\theta_1 - \theta_0)} \\ T &= \frac{D_* t}{{r_0}^2} \\ I &= \frac{i}{r_0(\theta_1 - \theta_0)} \end{aligned} \tag{136}$$

It follows from Eq. (69) that

$$m = 2, \qquad T = \pi t/t_{\text{geom}}; \qquad m = 3, \qquad T = \pi t/4t_{\text{geom}} \tag{137}$$

For $m = 2$, the solution of Eq. (134) subject to conditions (135) [23, p. 335] is complicated. It suffices here to give the result for the absorption rate

$$V = \frac{4}{\pi^2}\int_0^\infty \frac{\exp(-w^2 T)\,dw}{w[\mathbf{J}_0^2(w) + \mathbf{Y}_0^2(w)]} \tag{138}$$

$\mathbf{J}_0$ and $\mathbf{Y}_0$ are the Bessel functions of the first and second kind of zero order. Numerical values of the integral are given by Jaeger and Clarke [112] and are graphed by Carslaw and Jaeger [23, p. 338]. For small T, the series expansion of Eq. (138)

$$V = (\pi T)^{-1/2} + \tfrac{1}{2} - \tfrac{1}{4}(T/\pi)^{1/2} + (T/8) - \cdots \tag{139}$$

is useful. The asymptotic expansion (large T) is

$$V \sim 2\left[\frac{1}{\log(4T) - 2\gamma} - \frac{\gamma}{[\log(4T) - 2\gamma]^2} + \cdots\right] \tag{140}$$

$\gamma = 0.57722 \cdots$ is Euler's constant.

For $m = 3$, the solution of Eq. (134) subject to conditions (135) [23, p. 247] is

$$\vartheta = \theta_0 + \frac{\theta_1 - \theta_0}{\rho} \operatorname{erfc} \frac{\rho - 1}{2T^{1/2}} \tag{141}$$

This yields the simple result [88]

$$V = (\pi T)^{-1/2} + 1 \tag{142}$$

The leading terms of Eqs. (139) and (142) are identical; and the second term of Eq. (142) is exactly twice that of Eq. (139). The corresponding nonlinear results have the same features.

C. Infiltration Solutions

1. *One-Dimensional Infiltration*

The linearized form of Eqs. (81) and (82) is

$$\frac{\partial \vartheta}{\partial t} = D_* \frac{\partial^2 \vartheta}{\partial z^2} - k \frac{\partial \vartheta}{\partial z} \tag{143}$$

$$\begin{aligned} t &= 0, \quad z > 0, \quad \vartheta = \theta_0 \\ t &\geqq 0, \quad z = 0, \quad \vartheta = \theta_1 \end{aligned} \tag{144}$$

We have fixed D_* already through Eq. (132). We now, similarly, fix k by matching values of $\lim_{t\to\infty} v_0$. We thus obtain the value

$$k = (K_1 - K_0)/(\theta_1 - \theta_0) \tag{145}$$

so that k is identical with u, the velocity of the "profile at infinity" [Eq. (106)].

The solution of Eq. (143) subject to conditions (144) is [87]

$$\vartheta = \theta_0 + \frac{\theta_1 - \theta_0}{2}\left[\operatorname{erfc}\frac{z - kt}{2(D_* t)^{1/2}} + \exp\left(\frac{kz}{D_*}\right)\operatorname{erfc}\frac{z + kt}{2(D_* t)^{1/2}}\right] \tag{146}$$

As $t \to \infty$ in Eq. (146), the solution (at other than small z) approaches the form

$$z(\vartheta, t) = kt + t^{1/2}\omega(\vartheta) \tag{147}$$

The centroid of the profile tends to move downward with constant velocity $k(= u)$, and the profile tends to maintain its relative shape about the centroid, with the gradients of ϑ decreasing as $t^{-1/2}$. It is illuminating to compare Eq. (147) with the corresponding nonlinear result, Eq. (103). The constant absolute shape of the "profile at infinity," Eq. (103), is a demonstrable consequence of nonlinearity in $K(\theta)$.

We express the dynamics of one-dimensional infiltration in terms of the following dimensionless variables:

$$\begin{aligned} V_1 &= \frac{v_0 - K_1}{K_1 - K_0}, \qquad T_1 = \frac{k^2 t}{4D_*} = \frac{(K_1 - K_0)^2 t}{\pi S^2} = \frac{t}{\pi t_{\text{grav}}} \\ I_1 &= \int_0^{T_1} V\, dT' = \frac{K_1 - K_0}{\pi S^2}(i - K_1 t) \end{aligned} \tag{148}$$

Then, under the simplifying, and generally justified, assumption that K_0/K_1 is negligibly small, it follows from Eq. (146) that

$$\begin{aligned} V_1 &= \tfrac{1}{2}[(\pi T_1)^{-1/2}\exp(-T_1) - \operatorname{erfc} T_1^{1/2}] \\ I_1 &= \tfrac{1}{2}[\pi^{-1/2}T_1^{1/2}\exp(-T_1) + \tfrac{1}{2}\operatorname{erf} T_1^{1/2} - T_1 \operatorname{erfc} T_1^{1/2}] \end{aligned} \tag{149}$$

The series expansions of Eq. (149), useful for small T_1, are

$$\begin{aligned} V_1 &= \tfrac{1}{2}\left[(\pi T_1)^{-1/2} - 1 + \pi^{-1/2}\left\{T_1^{1/2} - \frac{T_1^{3/2}}{6} + \frac{T_1^{5/2}}{30} - \frac{T_1^{7/2}}{168} + \cdots\right\}\right] \\ I_1 &= \pi^{-1/2}T_1^{1/2} - \frac{T_1}{2} + \pi^{-1/2}\left\{\frac{T_1^{3/2}}{3} - \frac{T_1^{5/2}}{30} + \frac{T_1^{7/2}}{210} - \frac{T_1^{9/2}}{1512} + \cdots\right\} \end{aligned} \tag{150}$$

The asymptotic expansions for large T_1 are

$$\begin{aligned} V_1 &\sim \tfrac{1}{4}\pi^{-1/2}T_1^{-3/2}\exp(-T_1)\left[1 - \frac{3}{4T_1} + \frac{15}{4T_1{}^2} - \cdots\right] \\ I_1 &\sim \tfrac{1}{4}\left[1 - \pi^{-1/2}T_1^{-3/2}\exp(-T_1)\left\{1 - \frac{3}{T_1} + \cdots\right\}\right] \end{aligned} \tag{151}$$

2. Two- and Three-Dimensional Infiltration

The linearized formalism describing infiltration from the surfaces of a cylindrical ($m = 2$) or spherical ($m = 3$) cavity consists of Eq. (125) subject to the conditions (135).

The substitution [87]

$$\vartheta_* = (\vartheta - \theta_0) \exp\left[-\frac{kz}{2D_*} + \frac{k^2 t}{4D_*} \right] \tag{152}$$

reduces Eq. (125) to

$$\partial\vartheta_*/\partial t = D_* \nabla^2 \vartheta_* \tag{153}$$

Now, *so long as $kr_0/2D_*$ is small,* we may replace conditions (135) by

$$\begin{aligned} t &= 0, \qquad \vartheta_* = 0, && r > r_0 \\ t &\geqq 0, \qquad \vartheta_* = (\theta_1 - \theta_0) \exp \frac{k^2 t}{4D_*}, && r = r_0 \end{aligned} \tag{154}$$

We thus reduce these problems to ones of diffusion in cylindrical and spherically symmetrical systems with a boundary concentration increasing exponentially with time.

We discuss the solution in terms of T_1 and the new dimensionless quantities

$$R_* = \frac{kr}{4D_*}, \qquad R_0 = \frac{kr_0}{4D_*}, \qquad \sin\psi = \frac{z}{r}, \qquad V_2 = \frac{\bar{v}_0}{k(\theta_1 - \theta_0)} \tag{155}$$

$\bar{v}_0$ is the average value of v_0 over the supply surface. The solutions we present appear to be useful for $R_0 \leqq 0.25$, a range that includes many practical situations. The expressions for V_2 given below, and later, are simplified through the (usually justifiable) assumption that K_0/K_1 is negligibly small.

For $m = 2$, the solution of Eq. (153) subject to conditions (154) involves some complicated integrals. In the limit as $T_1 \to \infty$, it is, for R_0 small,

$$\vartheta = \theta_0 + (\theta_1 - \theta_0) \exp[2(R_* - R_0) \sin\psi] \frac{\mathbf{K}_0(2R_*)}{\mathbf{K}_0(2R_0)} \tag{156}$$

With suitable changes of symbolism, Eq. (156) for R_0 small is of the same form as Eq. (117), as it should be. Equation (156) gives, for R_0 small, the final infiltration rate

$$V_2(\infty) = [4R_0 \mathbf{K}_0(2R_0)]^{-1} \tag{157}$$

By averaging v_0 over only the *lower semicircle*, we may use Eq. (156) to obtain a first estimate (demonstrably an *underestimate*) of the final infiltration rate from a semicircular furrow. We thus obtain

$$V_2(\infty) = \pi^{-1} + [4R_0 \mathbf{K}_0(2R_0)]^{-1} \tag{158}$$

For $m = 3$, the solution of Eq. (153) subject to conditions (154) reduces, for R_0 small, to

$$\vartheta = \theta_0 + \frac{(\theta_1 - \theta_0)R_0}{2R_*} \exp[2(R_* - R_0)\sin\psi]$$
$$\times \left[\exp\{-2(R_* - R_0)\} \operatorname{erfc} \frac{R_* - R_0 - T_1}{T_1^{1/2}} \right.$$
$$\left. + \exp\{2(R_* - R_0)\} \operatorname{erfc} \frac{R_* - R_0 + T_1}{T_1^{1/2}} \right] \tag{159}$$

In the limit as $T_1 \to \infty$, Eq. (159) becomes

$$\vartheta = \theta_0 + \frac{(\theta_1 - \theta_0)R_0}{R_*} \exp[-2(R_* - R_0)(1 - \sin\psi)] \tag{160}$$

With the symbolism appropriately adjusted, Eq. (160) for R_0 small is of the same form as Eq. (122), as it should be. Equation (159) yields the infiltration rate

$$V_2 = \tfrac{1}{2}[(\pi T_1)^{-1/2} \exp(-T_1) + (2R_0)^{-1} + \operatorname{erf} T_1^{1/2}] \tag{161}$$

and

$$V_2(\infty) = \tfrac{1}{2} + (4R_0)^{-1} \tag{162}$$

Averaging v_0 over the lower hemisphere, we obtain the following (lower) estimates for infiltration from a hemispherical basin:

$$V_2 = \tfrac{1}{2}[(\pi T_1)^{-1/2}\exp(-T_1) + \tfrac{1}{2} + (2R_0)^{-1} + \operatorname{erf} T_1^{1/2}] \tag{163}$$

$$V_2(\infty) = \tfrac{3}{4} + (4R_0)^{-1} \tag{164}$$

D. Large-Time Solutions in Two and Three Dimensions

For two- and three-dimensional absorption problems that approach a steady state, the linearized Eq. (125) reduces, as $t \to \infty$, to

$$\nabla^2\vartheta = 0 \tag{165}$$

which is identical in form to Eq. (74). Linearization solutions for this class of problem are thus exact in the limit as $t \to \infty$, when the one-to-one relation between ϑ and θ is

$$\vartheta = \theta_0 + (\theta_1 - \theta_0) \int_{\theta_0}^{\theta} D\, d\theta \bigg/ \int_{\theta_0}^{\theta_1} D\, d\theta \tag{166}$$

For infiltration problems of this type, Eq. (125) becomes, in the limit as $t \to \infty$,

$$\nabla^2\vartheta = \frac{k}{D_*}\frac{\partial\vartheta}{\partial z} \tag{167}$$

This is identical in form to Eq. (114), so long as

$$\alpha = k/D_* \tag{168}$$

Linearization solutions of these problems are exact in the limit as $t \to \infty$ for media and soils satisfying Eq. (113). In such cases, also, Eq. (166) is the one-to-one relation between ϑ and θ in the limit as $t \to \infty$.

These results suggest that linearization should be a fairly reliable technique when applied to two- and three-dimensional problems at large t.

VII. Delta-Function Solutions

A. Principles of Method

Green and Ampt [113] were the first to propose a model of water movement during absorption and infiltration from free water (or water under positive hydrostatic pressure) which involved a definite sharp wet front, behind which θ and K are supposed to take the constant values $\bar{\theta}$ and $\bar{K}$, and ahead of which θ retains its initial value θ_0. The influence of capillarity (and of any constant nonzero hydrostatic pressure of the supplied water) is modeled by a constant difference in moisture potential between the supply surface and the wetting front, for which we shall use the symbol C with the convention that C is positive.

For systems involving only one space coordinate (one-dimensional, and cylindrically and spherically symmetrical systems) the resulting formalism replaces the basic partial differential equation considered in Sects. II–V by an ordinary differential equation in i and t. The parameters $\bar{\theta}$, $\bar{K}$, C enter equations for absorption only in the combination $\bar{K}C(\bar{\theta} - \theta_0)$; for infiltration, $\bar{K}$ also enters separately.

Lambe [114] recognized that, in general, $\bar{\theta} < \theta_1$ and $\bar{K} < K_1$. Philip [115] put the model on a somewhat firmer basis by showing that, in one-dimensional systems, the formalism gave $i(t)$ exactly, provided the moisture profiles, whatever their shape, preserved similarity. He showed that, under these conditions, the appropriate values of $\bar{\theta}$ and $\bar{K}$ are

$$\bar{\theta} = \int_0^{\eta_{wf}} \theta \, d\eta / \eta_{wf}, \qquad \bar{K} = v_0 \eta_{wf} \Big/ \int_0^{\eta_{wf}} v/K \, d\eta \tag{169}$$

η is the space coordinate, with $\eta = 0$ at the supply surface and $\eta = \eta_{wf}$ at the wet front. v is the flow velocity.

Subsequently, Philip [84] related the model to the "diffusion" description of absorption and infiltration described in Sects. II–V by showing that the model is exact for a medium in which the moisture diffusivity $D(\theta)$ is defined as follows:

$$D = S^2(\theta_1 - \theta_0)\delta(\theta_1 - \theta)/2 \tag{170}$$

δ is the Dirac delta function. Equation (170) signifies that D is very large in a very small θ region near $\theta = \theta_1$, is negligibly small in the rest of the θ range, and is such that $\int_{\theta_0}^{\theta_1} D\, d\theta = S^2/2$. The model does not require any particular form of $K(\theta)$, although it is physically evident that media tending to satisfy Eq. (170) must also have a very sharp peak in K at $\theta = \theta_1$. The gradual steepening of moisture profiles as the peak in D becomes sharper is shown by Philip [48]. The delta-function medium embodies, essentially, the limiting result of this process. With the model interpreted thus, $\bar{\theta} = \theta_1$.

The similarity condition for the validity of the model in one-dimensional systems is rather less restrictive than Eq. (170); but the model can be given a precise meaning in two- and three-dimensional systems [for which moisture profiles in media not satisfying Eq. (170) do not satisfy similarity even approximately] only through Eq. (170).

So far as soils of arbitrary hydrological characteristics are concerned, delta-function solutions give no information about details of the moisture profiles, but they do offer estimates of integral properties such as $v_0(t)$ and $i(t)$. Just as for the linearized model, we may use this model also as a means of extrapolation and interpolation. We fix $\bar{K}C(\theta_1 - \theta_0)$ by matching known integral properties of the one-dimensional absorption solution [86, 111]; and we assign $\bar{K}$ in problems involving gravity by matching known behavior as $t \to \infty$ [74, 111].

B. Absorption Solutions

1. One-Dimensional Absorption

The delta-function equation for one-dimensional absorption is

$$v_0 = \frac{di}{dt} = \frac{\bar{K}C(\theta_1 - \theta_0)}{i} \tag{171}$$

subject to the condition

$$t = 0, \qquad i = 0 \tag{172}$$

The solution is

$$i = [2\bar{K}C(\theta_1 - \theta_0)]^{1/2} t^{1/2} \tag{173}$$

Matching this with Eq. (42), we have

$$\bar{K}C(\theta_1 - \theta_0) = S^2/2 \tag{174}$$

Use of Eq. (174) for media of arbitrary properties ensures that delta-function solutions yield correct integral behavior, in the limit as $t \to 0$, for *all* absorption and infiltration processes induced by a step-function change in θ (or Ψ) at the supply surface.

2. *Two- and Three-Dimensional Absorption*

Substituting Eqs. (174) and (132) in Eq. (136), we see that we may use the dimensionless variables V, T, I to present delta-function solutions for the two- and three-dimensional absorption processes for which we have already considered the exact and linearized solutions.

For the two-dimensional case, the delta-function model yields the differential equation

$$V = dI/dT = (4/\pi)[\log(1 + 2I)]^{-1} \tag{175}$$

subject to the condition

$$T = 0, \qquad I = 0 \tag{176}$$

Integration gives

$$T = (\pi/8)[(1 + 2I)\log(1 + 2I) - 2I] \tag{177}$$

Equations (175) and (177) may then be combined to give the relation between V and T:

$$T = \frac{\pi}{8}\left[\left(\frac{4}{\pi V} - 1\right)e^{4/\pi V} + 1\right] \tag{178}$$

In three dimensions, the differential equation is

$$V = dI/dT = (2/\pi)[1 - (1 + 3I)^{-1/3}]^{-1} \tag{179}$$

subject to condition (176). Integration yields

$$T = (\pi/2)[I - \tfrac{1}{2}\{(1 + 3I)^{2/3} - 1\}] \tag{180}$$

Combining Eqs. (179) and (180) gives the relation between V and T:

$$T = \frac{\pi}{12}\left[2\left(1 - \frac{2}{\pi V}\right)^{-3} - 3\left(1 - \frac{2}{\pi V}\right)^{-2} + 1\right] \tag{181}$$

C. Infiltration Solutions

1. *One-Dimensional Infiltration*

The delta-function equation for one-dimensional infiltration is

$$v_0 = \frac{di}{dt} = \bar{K}\left(1 + \frac{C(\theta_1 - \theta_0)}{i}\right) \tag{182}$$

According to Eq. (182),

$$v_0(\infty) = \bar{K} \tag{183}$$

so that, matching this with Eq. (109), we find

$$\bar{K} = K_1 \tag{184}$$

We therefore use matching equation (184), as well as Eq. (174), in the delta-function model of infiltration. The model thus agrees with integral properties of the exact solution both as $t \to 0$ and as $t \to \infty$.

Substituting Eqs. (174) and (184) (and using the simplifying, but generally justified, assumption that K_0/K_1 is negligibly small) in Eq. (148), we find that we may use the dimensionless variables V_1, T_1, I_1, to discuss the delta-function solution for one-dimensional infiltration. In terms of these variables, Eq. (182) becomes

$$V_1 = \frac{dI_1}{dT_1} = \frac{1}{2\pi(I_1 + T_1)} \tag{185}$$

subject to condition

$$T_1 = 0, \qquad I_1 = 0 \tag{186}$$

The solution is

$$T_1 = (2\pi)^{-1}[\exp(2\pi I_1) - 2\pi I_1 - 1] \tag{187}$$

Combining Eqs. (185) and (187) gives T_1 (V_1):

$$T_1 = (2\pi)^{-1}[V_1^{-1} - \log(1 + V_1^{-1})] \tag{188}$$

2. *Two- and Three-Dimensional Infiltration*

For two- and three-dimensional infiltration, the delta-function model does not lead to an ordinary equation, but to a complicated formulation in which the wet front is a moving "free surface." The delta-function model appears, in any case, to be of doubtful accuracy for two- and three-dimensional processes, particularly for large t, since it fails to embody two important properties that may be established from exact solutions and from general physical arguments (and, somewhat more circumstantially, from linearized solutions). These are (i) that, for such processes, moisture profiles fail to preserve similarity and become progressively less steep as t increases; (ii) that, in systems with a steady state, the region of significant increase of moisture content (i.e., the region behind any "wet front") *remains definitely finite in both horizontal and vertical directions as* $t \to \infty$.

Vedernikov and Averjanov [cf. 107] have used delta-function models to study two-dimensional steady infiltration, but the results may be of limited value because of these physical shortcomings of the model.

VIII. Comparison Technique for Estimating Integral Properties of Solutions

A. Principles of Method

Figure 12 compares schematically the forms of $D(\theta)$ dependence for linearized and delta-function media with that for a typical real soil or porous

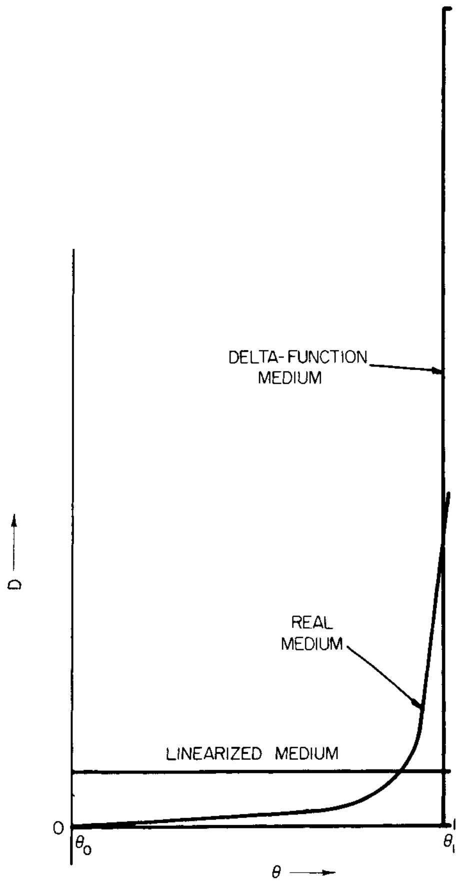


FIG. 12. Schematic figure comparing mode of variation of D with θ for (i) the delta-function medium, (ii) the linearized medium, and (iii) a typical real soil or porous medium.

medium. The latter, with a marked decrease of D with θ, is evidently intermediate in character between that of the two media treated in the preceding two sections.

We therefore expect integral forms of exact solutions for absorption processes [which depend only on $D(\theta)$] in real media to be intermediate in character between those given by the (appropriately matched) linearized and delta-function solutions for the same process [86, 91–93, 111]. The linearized and delta-function solutions thus form *an envelope of possible behavior.*

For processes involving gravity, however, solutions are influenced, not only by $D(\theta)$, but also by the relative behavior of $D(\theta)$ and $dK/d\theta$. Both linearized and delta-function models are consistent with $D^{-1}\, dK/d\theta$ being constant, whereas, for media of arbitrary characteristics, this is not necessarily so. It follows that we cannot extend the envelope concept to such processes: we limit ourselves to the expectation that the difference between linear and delta-function solutions will indicate the order of magnitude of the deviation of either solution from the exact result for any real medium.

These notions have been confirmed by comparing exact nonlinear solutions for particular real media with linear and delta-function solutions. Some details are given below. The various differences between solutions are small, so that we have at our disposal a useful means of estimating integral behavior of media of arbitrary characteristics.

We are able to employ this approach at three different levels:

(i) For absorption phenomena with both linearized and delta-function solutions available, a close estimate can be made of the integral behavior of any real medium (and the possible error of the estimate is known).

(ii) For infiltration phenomena with both linearized and delta-function solutions available, either gives an estimate of the integral behavior of any real medium. An estimate of the order of magnitude of the error is provided by the difference between the two solutions.

(iii) For phenomena where only one form of solution (usually the linearized solution) is available, that solution offers a fairly reliable estimate of integral behavior.

B. Absorption Solutions

1. *Two-Dimensional Absorption*

Figure 13 compares $V(T)$ for two-dimensional absorption according to the linearized solution, the delta-function solution, and the nonlinear solution (68) (first two terms) calculated for our illustrative example of Yolo light clay. The three solutions evidently agree very closely. Consonant with the concepts developed in Sect. VIII.A, the linearized and delta-function solutions indicate upper and lower bounds to the behavior of real media.

We compare the leading terms of the various $V(T)$ expansions appropriate for small T:

$$\begin{array}{lll} \text{linearized solution, Eq. (139),} & V = (\pi T)^{-1/2} + \frac{1}{2} - \cdots & \\ \text{delta-function solution, Eq. (178),} & V = (\pi T)^{-1/2} + (4/3\pi) - \cdots & (189) \\ \text{nonlinear example, Eq. (68),} & V = (\pi T)^{-1/2} + 0.44 - \cdots & \\ 4/3\pi = 0.424 \ldots . & & \end{array}$$

The comparison in the limit of T large is as follows:

$$\begin{array}{lll} \text{linearized solution, Eq. (140),} & V \sim 2(\log T)^{-1} & \\ \text{delta-function solution, Eq. (178),} & V \sim (4/\pi)(\log T)^{-1} & (190) \end{array}$$

The results of Eqs. (190) establish that, for media of arbitrary characteristics, the asymptotic behavior of V at large T is as $(\log T)^{-1}$. For real media, the numerical factor will be between the limits $4/\pi$ and 2.

2. *Three-Dimensional Absorption*

Figure 14 shows the similar comparison of $V(T)$ calculations for three-dimensional absorption. In this case, the comparison includes also the exact nonlinear solution for $V(\infty)$ [Eq. (71)] computed for our illustrative example. Again the solutions are in good agreement, and are consistent with the notion of the linearized and delta-function solutions as upper and lower envelopes of the possible behavior of real media.

The leading terms of the various expansions for $V(T)$ at small T are compared in Eqs. (191):

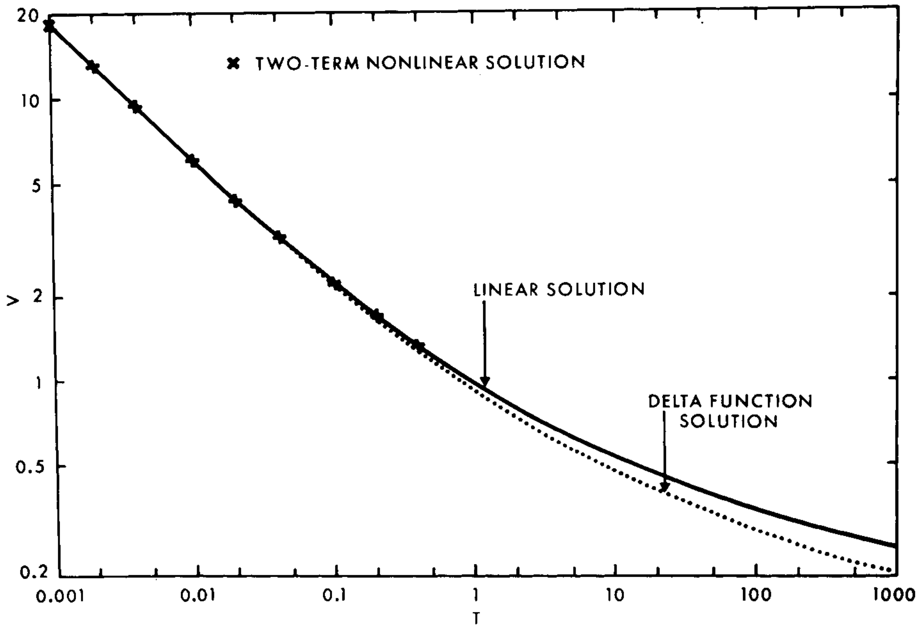
$$\begin{array}{lll} \text{linearized solution, Eq. (142),} & V = (\pi T)^{-1/2} + 1 & \\ \text{delta-function solution, Eq. (181),} & V = (\pi T)^{-1/2} + (8/3\pi) - \cdots & (191) \\ \text{nonlinear example, Eq. (68),} & V = (\pi T)^{-1/2} + 0.87 - \cdots & \\ 8/3\pi = 0.848 \ldots . & & \end{array}$$

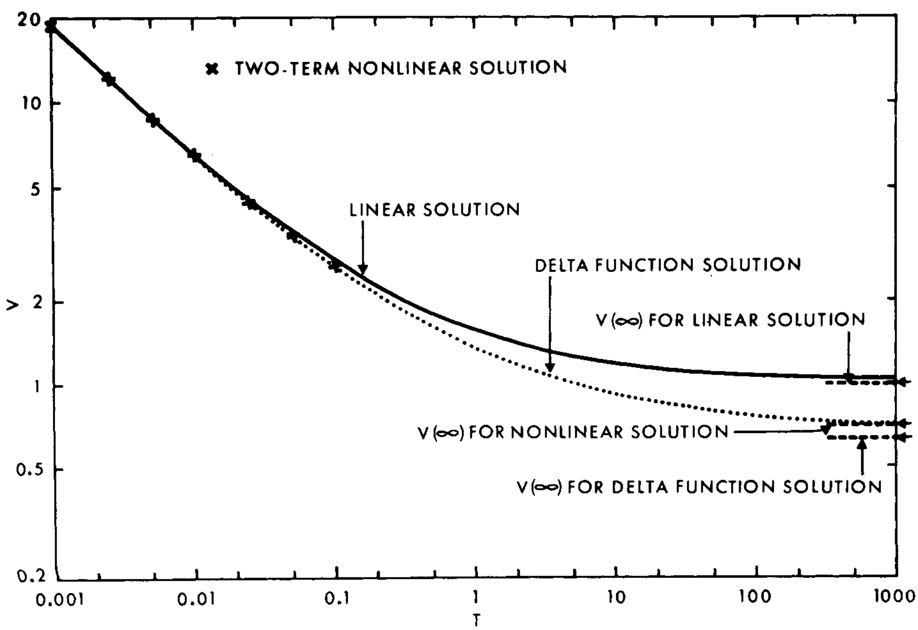
In the limit of T large, the comparison is

$$\begin{array}{lll} \text{linearized solution, Eq. (142),} & V = 1 + (\pi T)^{-1/2} & \\ \text{delta-function solution, Eq. (181),} & V \sim (2/\pi) + (4\pi^2/3)^{1/3} T^{-1/3} & (192) \\ \text{nonlinear example, Eq. (71),} & V \sim 0.710 & \\ 2/\pi \approx 0.636 \ldots . & & \end{array}$$

FIG. 13. The dynamics of two-dimensional absorption in the dimensionless form $V(T)$ [86]. Solid curve: the linearized solution Eq. (139). Crosses: two-term nonlinear solution for Yolo light clay Eq. (68). Dotted curve: delta-function solution Eq. (178).

FIG. 14. The dynamics of three-dimensional absorption in the dimensionless form $V(T)$ [86]. Solid curve: the linearized solution Eq. (142). Crosses: two-term nonlinear solution for Yolo light clay Eq. (68). Dotted curve: delta-function solution Eq. (181). Values of $V(\infty)$ are shown for all three solutions. Nonlinear $V(\infty)$ solution calculated from Eq. (71).


TWO-TERM NONLINEAR SOLUTION
LINEAR SOLUTION
DELTA FUNCTION SOLUTION
V
T
20
10
5
2
1
0.5
0.2
0.001
0.01
0.1
1
10
100
1000


TWO-TERM NONLINEAR SOLUTION
LINEAR SOLUTION
DELTA FUNCTION SOLUTION
V(∞) FOR LINEAR SOLUTION
V(∞) FOR NONLINEAR SOLUTION
V(∞) FOR DELTA FUNCTION SOLUTION
V
T
20
10
5
2
1
0.5
0.2
0.001
0.01
0.1
1
10
100
1000

C. Infiltration Solutions

The various solutions for one-dimensional capillary rise [76] and infiltration also agree closely, although the "envelope" concept cannot be extended to these processes involving gravity. Figure 15 compares various $V_1(T_1)$ calculations for one-dimensional infiltration. The nonlinear solution (97) for our illustrative example was calculated from the first four terms.

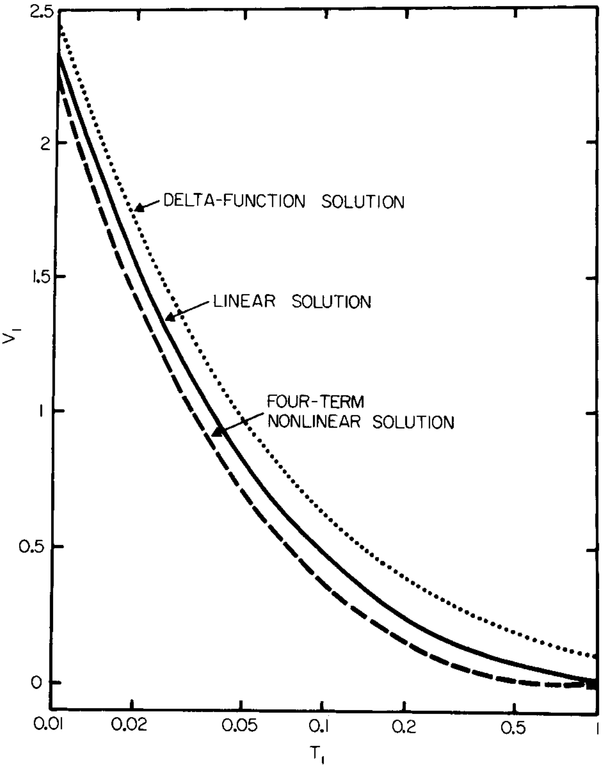


FIG. 15. The dynamics of one-dimensional infiltration in the dimensionless form $V_1(T_1)$ [111]. Solid curve: the linearized solution Eq. (150). Broken curve: nonlinear four-term solution for Yolo light clay Eq. (97). Dotted curve: delta-function solution Eq. (188).

The leading terms of the various $V_1(T_1)$ expansions for small T_1 are

linearized solution, Eq. (150), $\quad V_1 = \frac{1}{2}(\pi T_1)^{-1/2} - \frac{1}{2} + \cdots$

delta-function solution, Eq. (188), $\quad V_1 = \frac{1}{2}(\pi T_1)^{-1/2} - \frac{1}{3} + \cdots$ [84] $\quad$ (193)

nonlinear example, Eq. (97), $\quad V_1 = \frac{1}{2}(\pi T_1)^{-1/2} - 0.62 + \cdots$

For $T_1 \to \infty$, the asymptotic results are

$$\begin{array}{lll} \text{linearized solution, Eq. (151),} & V_1 \sim \frac{1}{4}\pi^{-1/2}T_1^{-3/2}\exp(-T_1) & \\ \text{delta-function solution, Eq. (188),} & V_1 \sim (2\pi T_1)^{-1} & (194) \\ \text{nonlinear example, Eq. (109),} & V_1 \sim 0 & \end{array}$$

The nonlinear analysis of Sect. V.B is unable to give details of the approach of V_1 to 0 (i.e., of v_0 to K_1). Since the linearized model does not embody the "profile at infinity" behavior (Sect. VI.C), whereas the delta-function model does, the approach to the quasi-stationary state with $V_1 = 0$ (i.e., $v_0 = K_1$) for real media may resemble the T_1^{-1} behavior of the delta-function model, rather than the more rapid approach of the linearized model.

On the basis of the analysis of Sect. V.A, Philip [33, 84] suggested the two-parameter infiltration equations

$$\begin{aligned} i &= St^{1/2} + At \\ v_0 &= \tfrac{1}{2}St^{-1/2} + A \end{aligned} \tag{195}$$

for use in applied hydrology when t is not too large. Clearly the relation

$$v_0 = K_1 \qquad (196) \text{ [also Eq. (109)]}$$

holds in the limit as $t \to \infty$.

It has been suggested recently [19] that this implies that A must be K_1. However, *the coefficients of corresponding terms in a series expansion with limited radius of convergence and in an asymptotic (large argument) expansion of the same function are not necessarily equal.* We have, in fact, from Eqs. (193) that $A = K_1/2$, $2K_1/3$, and $0.38K_1$ for the linearized model, the delta-function model, and for our nonlinear example.

Only linearized solutions are presently available for transient two- and three-dimensional infiltration. The comparison technique is, so far, useful for these problems only in the sense that it indicates that the process of linearization, combined with matching, yields results of useful accuracy.

D. Aggregated Media

A comparison technique was applied recently to the problem of absorption into aggregated media. The study used linearized and wet-front (modified delta-function) models of water movement in the macroporosity [35, 36].

IX. Exact Solutions of More Complicated Problems

Sections IV–VIII have dealt exclusively with flow processes subject to concentration boundary conditions in homogeneous systems. Before proceeding in the final sections to a discussion of the physical significance of the solutions of Sects. IV–VIII, we examine briefly the more difficult problems of flow processes subject to flux conditions and of flow in heterogeneous media. For these problems, we identify special subclasses for which exact solutions may be found by methods similar to those considered earlier in this article.

A. Processes Subject to Flux Boundary Conditions

The additional mathematical difficulty of transient flow processes subject to flux boundary conditions arises, essentially, because the θ range varies with t (so that θ cannot readily be made an independent variable). Purely numerical techniques, made less laborious by the use of high-speed computers, may be the only means of analyzing many such problems [116–118]. However, several subclasses of exact solutions are available. We discuss these below.

1. *One-Dimensional Absorption*

One-dimensional absorption with constant surface flux into a semi-infinite homogeneous medium of uniform initial moisture content θ_0 is described by Eq. (15) subject to conditions (197):

$$\begin{aligned} t &= 0, \qquad x > 0, \qquad \theta = \theta_0 \\ t &\geqq 0, \qquad x = 0, \qquad D\,\partial\theta/\partial x = -U_0 \end{aligned} \tag{197}$$

When $D(\theta)$ may be represented by the form

$$D(\theta) = D_s\left(\frac{\theta - \theta_0}{\theta_s - \theta_0}\right)^n \qquad \text{for} \quad n > -1 \tag{198}$$

there exists a similarity solution of Eq. (15) subject to conditions (197). Here θ_s is the value of θ corresponding to $\Psi = 0$, and D_s denotes $D(\theta_s)$. In general, Eq. (198) offers a reasonably accurate representation, but an important limitation is that it requires $D(\theta_0) = 0$ for $n > 0$. For this reason, Eq. (198) is best fitted to soils and media that are initially rather dry. The appropriate range of n seems to be from about 3 to 7, although smaller values would apply if the method were used with large values of θ_0.

The dimensionless quantities

$$\Theta = \frac{\theta - \theta_0}{\theta_s - \theta_0}, \qquad \tau_1 = \frac{U_0{}^2 t}{D_s(\theta_s - \theta_0)^2}, \qquad \chi = \frac{U_0 x}{D_s(\theta_s - \theta_0)} \tag{199}$$

reduce Eq. (15) subject to conditions (197), with D defined by Eq. (198), to the form

$$\frac{\partial\Theta}{\partial\tau_1} = \frac{\partial}{\partial\chi}\left(\Theta^n \frac{\partial\Theta}{\partial\chi}\right) \tag{200}$$

subject to the conditions

$$\begin{aligned} \tau_1 &= 0, & \chi &> 0, & \Theta &= 0 \\ \tau_1 &\geqq 0, & \chi &= 0, & \Theta^n\, \partial\Theta/\partial\chi &= -1 \end{aligned} \tag{201}$$

The similarity substitutions

$$\Theta_* = \Theta\tau^{-1/(n+2)}, \qquad \chi_* = \chi\tau^{-(n+1)/(n+2)} \tag{202}$$

then enable the reduction of Eq. (200) subject to conditions (201) to an ordinary nonlinear differential equation in Θ_* and χ_*, subject to conditions expressed solely in terms of these quantities. Numerical methods yield the solution in the form $\Theta_*(\chi_*)$, i.e.,

$$\theta(x, t) = \theta_0 + t^{1/(n+2)}F(xt^{-(n+1)/(n+2)}) \tag{203}$$

Clearly, this solution is valid only so long as $\theta(0) \leqq \theta_s$, i.e., for

$$t \leqq \left(\frac{\theta_s - \theta_0}{F(0)}\right)^{n+2} \tag{204}$$

For larger t, the constant flux boundary condition can be maintained only by continuous increase of the hydrostatic pressure of the water supplied at $x = 0$.

2. *One-Dimensional Infiltration*

The preceding result is relevant also to the early stages of one-dimensional infiltration with a constant flux boundary condition. When $U_0 \leqq K_s$, we may supplement this result with an asymptotic large-time solution of the same character as that described in Sect. V.B, except that we now write U_0 for K_1 and θ_1 is replaced by θ_U, where $K(\theta_U) = U_0$.

For $U_0 > K_s$, the constant flux condition cannot be maintained indefinitely unless water is supplied to the surface under a continuously increasing hydrostatic pressure. The required rate of increase of pressure is, asymptotically,

$$\frac{(U_0 - K_0)(U_0 - K_s)}{K_s(\theta_s - \theta_0)} \quad \text{cm-sec}^{-1}$$

3. *Two- and Three-Dimensional Problems*

The techniques of Sects. IV.H and V.E for large-time (steady) absorption and infiltration in two and three dimensions apply for a flux condition at the supply surface as well as for a concentration condition. These techniques are appropriate only if U_0 is nowhere greater than the steady flux density on the supply surface when it is maintained at moisture content θ_s.

B. Absorption and Infiltration in Heterogeneous Media

1. *The Mathematical Problem*

For unsteady unsaturated water movement in the simplest heterogeneous system, namely, a horizontal one-dimensional one, Eq. (9) reduces to

$$\frac{\partial\theta}{\partial t} = \frac{\partial}{\partial x}\left[K(\theta, x)\frac{\partial}{\partial x}\Psi(\theta, x)\right] \tag{205}$$

Equation (205) implies that, for each point x, there is a definite functional dependence of K and Ψ on θ. Superficial consideration might suggest that, in analogy with the diffusion formalism for homogeneous media, Eq. (205) might reduce to a diffusion equation with the diffusivity a function of both θ and x. In fact, Eq. (205) reduces to the complicated and generally intractable form

$$\frac{\partial\theta}{\partial t} = \frac{\partial}{\partial x}\left[D(\theta, x)\frac{\partial\theta}{\partial x}\right] + E(\theta, x)\frac{\partial\theta}{\partial x} + F(\theta, x) \tag{206}$$

The difficulty of the problem is aggravated by the fact that there is little hope of receiving a realistic and useful picture of the properties of equations such as Eq. (205), so long as $K(\theta, x)$ and $\Psi(\theta, x)$ are free to vary totally independently of each other. The computer use of numerical methods (e.g., Hanks and Bowers [119]) is clearly helpful, but offers no panacea. The concept of scale-heterogeneity, referred to in Sect. IX.B.3, offers means of reducing the problem to more manageable proportions by using a physically based connection between the x variation of K and Ψ.

2. *Delta-Function Approach*

Childs [101] has recently carried further an early unpublished attempt by the present author to study one-dimensional infiltration in heterogeneous systems by a delta-function model. The model appears to be quite unsuited to profiles where the soil texture becomes coarser in depth. In this case, the lower parts of the profile will tend to be permanently at relatively large negative Ψ; moisture profiles in that region will tend to be gradual, unlike the delta-function model as used by Childs. The model, superficially, appears better fitted to the opposite case where coarse-textured soil overlies finer; but even here the model may lead to such paradoxes as an infiltration rate that increases when the putative wet front encounters the finer-textured material.

Further to these quite serious shortcomings, we observe that $\bar{K}$ and C must reflect in a mutually consistent way the heterogeneity of the medium. Manipulations in which these parameters vary with depth quite independently of each other [101] must, in this author's opinion, be treated with great caution.

3. Exact Solutions for Scale-Heterogeneous Media

Philip [120] has considered a limited class of heterogeneity called *scale-heterogeneity.* A scale-heterogeneous medium is one in which the internal geometry is everywhere geometrically similar, but in which the characteristic internal length λ is free to vary spatially. The properties of the medium are then defined by λ, and the reduced conductivity and potential functions $K_{**}(\theta)$ and $\Psi_{**}(\theta)$. In a one-dimensional system, for example,

$$\begin{aligned} K(\theta, x) &= [\lambda(x)]^2 \cdot K_{**}(\theta) \\ \Psi(\theta, x) &= \Psi_{**}(\theta)/\lambda(x) \end{aligned} \tag{207}$$

Equations (207) depend on the assumption that the flow process is essentially viscous flow of water of normal viscosity, and that Ψ is capillary in nature. See Miller and Miller [65] for a discussion of these concepts, and Klute and Wilkinson [121, 122] for experimental work verifying them.

Philip showed that for the special functional forms

$$\begin{aligned} K_{**} &\propto \Psi_{**}^{-2} \\ \Psi_{**} &\propto e^{-\beta\theta} \end{aligned} \tag{208}$$

Eq. (207) reduces Eq. (206) to a nonlinear diffusion equation in $\log(-\Psi)$. Eq. (208) at least embodies the general experience that K decreases rapidly as $|\Psi|$ increases; and it offers a range of modes of variation of Ψ with θ, the representation being exact where the *pF curve* is linear. An important limitation is that these relations cannot apply satisfactorily when Ψ exceeds the air-entry value anywhere in the medium.

The method of Sect. IV.A yields the solution of the problem of one-dimensional absorption into scale-heterogeneous media satisfying Eq. (208). The solution is obtained in terms of Ψ; the further calculation of the solution for θ is a simple matter. The approach has been extended to one-dimensional infiltration. An interesting aspect of this study is that the solution for Ψ is independent of the mode of variation of λ. This determines only the conversion of the solution for Ψ to the solution for θ.

X. Physics of One-Dimensional Infiltration

We return here to the implications of the exact solutions for one-dimensional infiltration into a homogeneous soil of uniform initial moisture content. These solutions constitute a striking example of interaction between capillary and gravity. As we have seen, capillary potential gradients are dominant at small t; and the final quasi-stationary state is an expression of dynamic equilibrium between capillary and gravitational influences, brought about by the nonlinearity of $K(\theta)$. The existence of relatively steep moisture gradients ("wetting zone" and "wet front") are manifestations of the nonlinearity of the flow equation: specifically, they arise whenever D decreases strongly with θ.

A. Moisture Profile

We have presented in Fig. 9 the sequence of moisture profiles calculated for our illustrative example. The salient features indicated by the solutions are as follows:

(i) For small t, the moisture profile maintains a nearly constant *relative* shape, z ordinates for fixed θ increasing at $t^{1/2}$. This is, of course, the phase when capillarity dominates, and the behavior is indistinguishable from that for one-dimensional absorption.

(ii) At intermediate t, the z ordinates gradually depart from their proportionality to $t^{1/2}$, and the profile shape changes slowly. The moisture gradient near $\theta = \theta_1$ becomes flatter, but it becomes *relatively* steeper over the rest of the θ range. The profile as a whole (except at $\theta = \theta_1$) moves down at a velocity that decreases continuously.

(iii) Finally, at very large t, these tendencies reach their full expression. The profile of nearly constant *absolute* shape attached to a lengthening near-straight close to $\theta = \theta_1$ moves down at a velocity indistinguishable from $(K_1 - K_0)/(\theta_1 - \theta_0)$.

B. Comparison with Experiment

The exact solutions for one-dimensional infiltration are in excellent agreement with the relevant experimental data [102, 123–130]. For work earlier than about 1957, the agreement cannot be confirmed in full detail, since, before then, the experimental soils were not, in general, characterized by their $D(\theta)$ and $K(\theta)$ functions. The later work tests the theory more rigorously.

The observed moisture profiles during infiltration are found to exhibit four zones, arranged as follows from the surface $z = 0$ downward:

(i) The saturated zone: a surface zone of presumed saturation extending (in experiments on infiltration from shallow ponded water) to a depth of the order of 1 cm.

(ii) The transmission zone: an upper region in which θ changes quite slowly with both z and t, which lengthens as infiltration proceeds.

(iii) The wetting zone: a region of fairly rapid change of θ with respect to both z and t.

(iv) The wet front: a region of very steep moisture gradient which represents the visible limit of moisture penetration.

These various zones are predicted correctly by the exact solutions. Philip [48] has given a detailed discussion of how the shape of the $D(\theta)$ function influences the shape of the moisture profiles, and so leads to the observed characteristics. The "saturated zone" is the region in which $\Psi \geqq \Psi_e$, and tends to become more significant as the water depth over the soil (the hydrostatic pressure at the supply surface) increases [22, 81].

Bodman and Colman [123, 124], and some later experimenters, observed a "transition zone" between the saturated zone and the transmission zone. This is a zone, up to a few centimeters deep, through which the moisture content decreases rapidly from "saturation" to a value characteristic of the transmission zone. It is noteworthy that this zone has been observed by some investigators, but not by others. An explanation in terms of air-entrapment has been offered, but it is possible that the transition zone is simply a

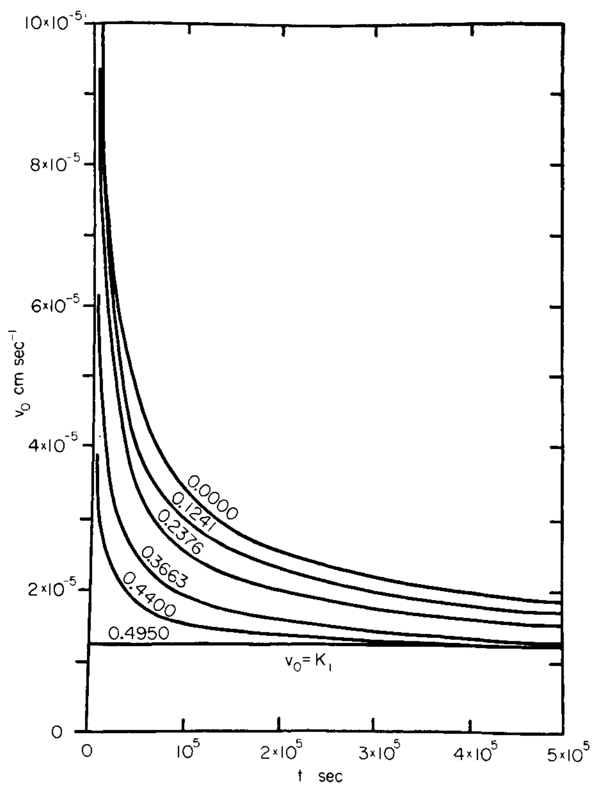


FIG. 16. Computed influence of the initial moisture content, θ_0, on one-dimensional infiltration rate v_0 for the soil of Figs. 1–3 with $\theta_1 = 0.4950$ [85]. Numerals on each curve denote values of θ_0. Note that curve for $\theta_0 = \theta_1$ is the line $v_0 = K_1$.

perturbation arising from anomalies of packing, or disturbances of structure, near the surface.

C. Effect of Initial Moisture Content

Figure 16 shows, for our illustrative example, the effect of the initial moisture content, θ_0, on the time-course of the infiltration rate, $v_0(t)$, as calculated from the exact solutions. Increasing θ_0 decreases v_0 at small t, but the effect decreases as t increases, and there is no influence on the final infiltration rate. On the other hand, increasing θ_0 *increases* the velocity of wet front advance at all t. These results are primarily the consequence of the decrease in "storage capacity" as θ_0 increases. The decrease of capillary potential differences is demonstrably a secondary influence. Philip [85] gives a detailed discussion of the influence of θ_0 on the shape and general behavior of the moisture profile.

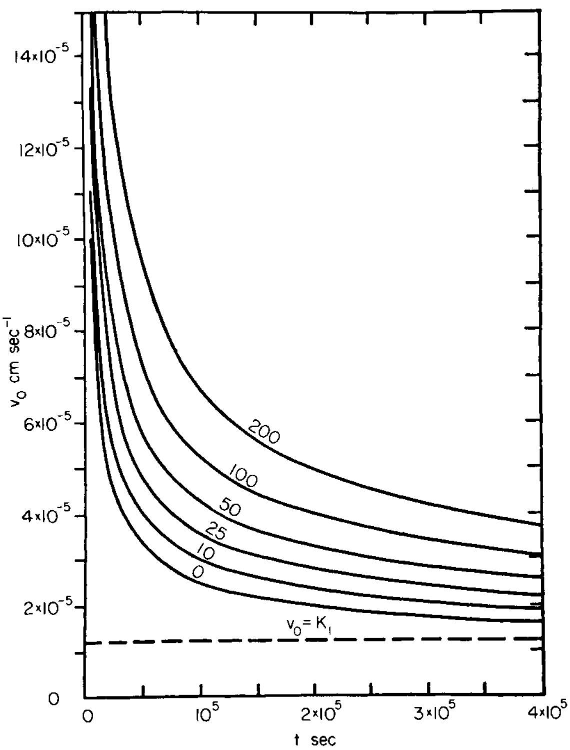


FIG. 17. Computed influence of water depth on one-dimensional infiltration rate v_0 for the soil of Figs. 1–3 with $\theta_0 = 0.2376$ [81]. Numerals on each curve denote values of water depth in centimeters. All curves are asymptotic to line $v_0 = K_1$.

D. Effect of Water Depth

Figure 17 presents, for our illustrative Yolo light clay, the effect of water depth over the soil on $v_0(t)$, as calculated from the exact solutions. Increasing the water depth increases v_0 at small t, but the effect becomes less important as t increases, and there is no effect in the limit as $t \to \infty$. The influence on wet front velocity is essentially similar. We have noticed already, in Sect. VII.A, that the parameter C of the delta-function model combines additively the magnitude of a putative wet front capillary potential and the hydrostatic pressure at the supply surface, so that C can be indifferently made up of various relative hydrostatic and capillary contributions. Exact analysis of the problem [81] shows that, in fact, the main effect of water depth on infiltration dynamics is that *an increase in water depth gives an increase in the sorptivity*, S. S is increased typically by about 2%/cm of water depth. The delta-function model is found to give a reliable prediction of the influence of water depth on S [22]. Increasing the water depth increases the steepness of moisture gradients in the wetting zone and the wet front (for small and moderate values of t) [81].

E. Algebraic Infiltration Equations

It is useful in applied hydrology to be able to characterize the dynamics of infiltration by a small number of parameters. Certain forms of algebraic infiltration equation cannot be related directly to the physical processes involved in infiltration. The Horton [131] exponential equation is particularly ill-fitted to describe the dynamics of one-dimensional infiltration into a uniform soil, despite its three disposable parameters, because it cannot adequately represent the $t^{1/2}$ behavior of i at small t. The Kostiakov [132]–Lewis [133] formulation is rather better, but, for infiltration into a uniform soil, its power law exponent for i must be $\frac{1}{2}$ at small t, and it must tend to increase gradually as data are fitted to larger t ranges.

The delta-function model yields an equation of the form

$$t = Y_1[i - Z_1 \log(1 + i/Z_1)] \tag{209}$$

where Y_1 and Z_1 have definite meanings in relation to either the similarity or delta-function interpretations of the model. [Despite its different appearance, Eq. (209) is exactly Eq. (187).] It will be evident from Sects. VIII and IX that Eq. (209) offers a good representation of one-dimensional infiltration into a uniform soil. Its implicit form is, however, inconvenient. It follows from Sects. VII and IX that the linearized solution offers an alternative two-parameter equation that will be equally reliable. Equations (149) are the $v_0(t)$ and $i(t)$ forms of this equation in dimensionless form. The linearized equation is at least explicit, but it seems far too complicated to be useful.

The representation

$$v_0 = \tfrac{1}{2}St^{-1/2} + A$$
$$i = St^{1/2} + At \tag{210}$$

based on the analysis of Sect. V.A, has the merit of being both physically well founded and simple. It appears to offer the most suitable two-parameter representation of the dynamics of infiltration. The limits on the validity of Eq. (210) at large t are discussed in Sect. IX.C. As we show there, when A is fixed by the behavior of v_0 or i at small t, it is definitely not equal to K_1. The form of Eq. (210) should, however, fit v_0 data over the whole t range fairly well, and a "best fit" over such a range will tend to give $A = K_1$. The matter of algebraic infiltration equations is discussed further by Philip [33, 84].

XI. Effects of Geometry and Gravity on Infiltration

The relevant two- and three-dimensional solutions of the flow equation lead to some fairly general insights into the interactions of capillarity, gravity, and geometry.

In the following discussion, two- (or three-) dimensional absorption and infiltration mean absorption and infiltration in a two- (or three-) dimensional region that is infinite (or semi-infinite) in the two (or three) dimensions from a supply surface that is finite in both (or all three) dimensions. The geometry we use for specific illustrations is as follows: Two-dimensional absorption: absorption in a cylindrically symmetrical infinite system from a supply surface of radius r_0. Three-dimensional absorption: absorption in a spherically symmetrical infinite system from a supply surface of radius r_0. Two-dimensional infiltration: infiltration in a two-dimensional system (infinite in the horizontal and semi-infinite in the vertical) from a semicircular furrow in its upper surface. Three-dimensional infiltration: infiltration into a three-dimensional region (infinite in both horizontal directions and semi-infinite in the vertical) from a hemispherical basin in its upper surface.

Strictly, t_{grav} will never be infinite, although it may be very large for some media. It follows that gravity must ultimately become nonnegligible in three-dimensional systems, even in the finest-textured media. Nevertheless, it remains useful and illuminating to include in our discussion (as we have done earlier) reference to the large t solutions for three-dimensional absorption.

A. Moisture Distribution

We have already observed that one-dimensional absorption moisture profiles preserve similarity. A well-defined wetting region and wet front are produced as a consequence of the shape of the $D(\theta)$ curve, but gradients (at fixed θ) everywhere decrease as $t^{-1/2}$, so that there is finally no real persistence of a wet front. In one-dimensional infiltration, on the other hand, nonlinearity in $K(\theta)$ produces the profile at infinity at large t, and a relatively sharp front persists for all t.

In two-dimensional absorption, profiles become less steep more rapidly than in one dimension, and gradients are always very small at large r. This can be regarded as a consequence of the fact that flow velocity must always decrease spatially more rapidly than r^{-1} in two-dimensional systems. In three-dimensional absorption, gradients decrease even more rapidly than in two dimensions: in this case, the flow velocity cannot decrease spatially less rapidly than r^{-2}. In three dimensions, however, there exists a final steady state, so that the gradients do not decrease to zero everywhere in the limit as $t \to \infty$, as they do in two dimensions.

There is no persistence of a wet front in two- and three-dimensional infiltration comparable to that in one dimension. In both cases, there is a final steady moisture distribution, and, everywhere within a finite distance of the supply surface, gradients tend to a small finite value.

The effect of gravity is, of course, to distort the symmetrical moisture distributions produced in the comparable symmetrical absorption problems. This distortion is very much stronger for two-dimensional systems than for three-dimensional ones [Figs. 9 and 10]. It also increases with increasing r_0 for both systems [86, 91]. The variation of the properties of large t moisture distributions during infiltration with the number of dimensions is a striking illustration of the influence of geometry on the effect of gravity. In one dimension, we have a quasi-stationary profile; in two dimensions, a very strongly distorted, but stationary, distribution; in three dimensions, a relatively weakly distorted, stationary distribution.

B. Existence of Steady State and Character of Ultimate Wetting

Table I summarizes the large-time behavior of these various systems. By *ultimate wetted region* (UWR), we mean the region that undergoes a definite, *nonzero and noninfinitesimal* increase of moisture content if the infiltration process is continued indefinitely. By *complete ultimate wetting*, we mean that, if infiltration continues indefinitely, all points at a finite distance from the supply surface reach a moisture content arbitrarily close to θ_1; by *incomplete wetting*, we mean that, even when infiltration continues indefinitely, all parts of the system a definite distance from the supply surface remain at moisture contents definitely less than that at the supply surface. Some modification of the *wetting* definitions is needed when the processes are complicated by

TABLE I

EFFECT OF GEOMETRY AND GRAVITY ON INFILTRATION

	Absorption (without gravity)			Infiltration (with gravity)		
No. of dimensions m	Ultimate wetted region	Ultimate wetting	Steady state?	Ultimate wetted region	Ultimate wetting	Steady state?
1	Infinite	Complete	No	Infinite	Complete	Quasi-steady
2	Infinite	Complete	No	Finite	Incomplete	Yes
3	Finite	Incomplete	Yes	Finite	Incomplete	Yes

air-entrapment. Also, where water is supplied at a potential greater than Ψ_e, it will be understood that there is a finite lower bound to the "definite distance from the supply surface" [22].

We note the following:

(i) The existence of a finite UWR, incomplete ultimate wetting, and a steady state are intimately connected. The one-dimensional gravity case is, in a sense, singular in that it has a quasi-steady state with an infinite UWR.

(ii) Gravity has a stabilizing effect. Two-dimensional UWR's are infinite without gravity, but are finite with gravity.

(iii) Increasing the number of dimensions has a stabilizing effect. No one-dimensional UWR's are finite; but two-dimensional UWR's with gravity are finite, and *all* three-dimensional UWR's are finite.

These considerations are obviously of great importance in connection with techniques of surface (and sub-) irrigation.

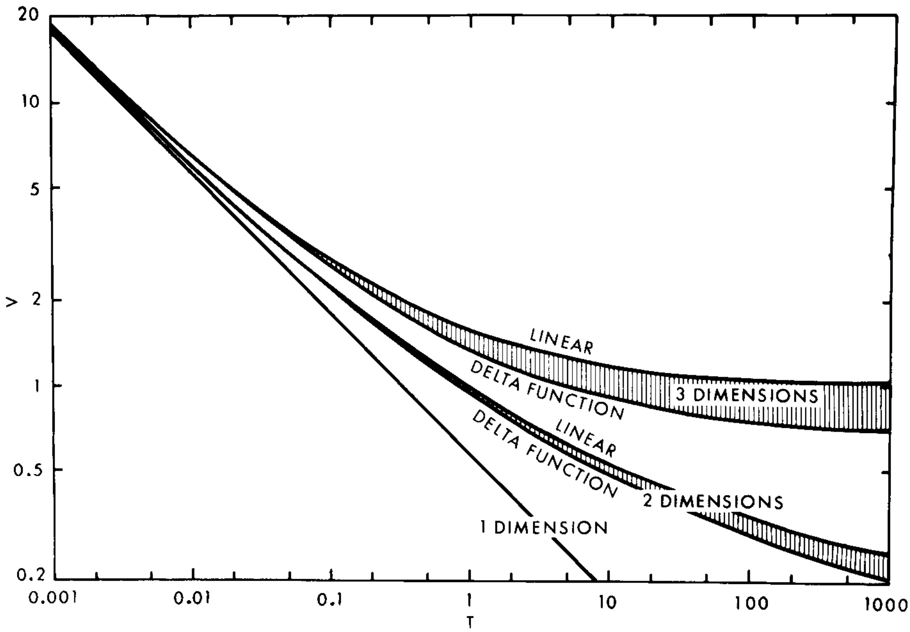
C. INFILTRATION RATE

The effects of geometry and gravity on the time course of the infiltration rate v_0 are illustrated by Figs. 18–20.

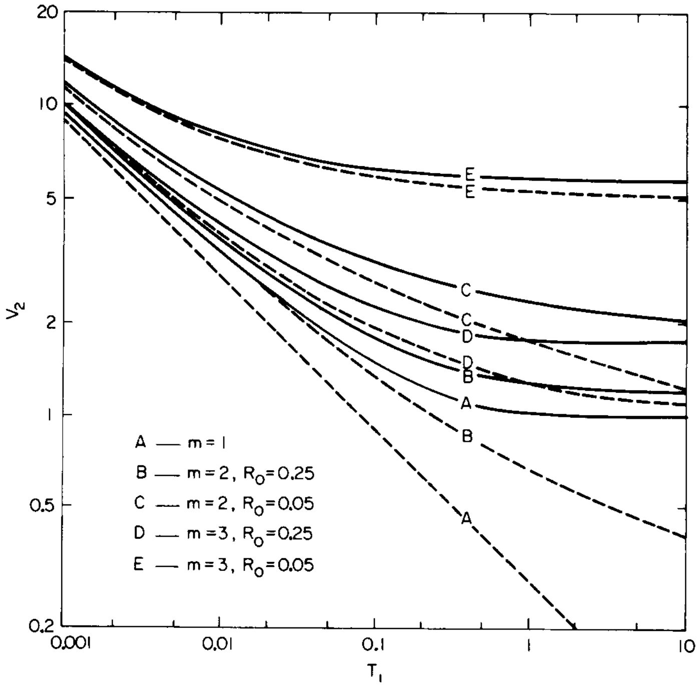
Figure 18 shows $V(T)$ for linearized and delta-function absorption solutions for $m = 1, 2$, and 3. Figure 19 brings together linearized results for

FIG. 18. The dynamics of absorption in one, two, and three dimensions in the dimensionless form $V(T)$ [86]. Shaded areas indicate range of variation of $V(T)$ as soil properties vary between the linearized and delta-function extremes.

FIG. 19. The dynamics of absorption and infiltration in one, two, and three dimensions (linearized solutions) in the dimensionless form $V_2(T_1)$ [86]. Solutions for $m = 2$ and 3 are shown for $R_0 = 0.05$ and 0.25. Full curves: infiltration. Broken curves: absorption.



LINEAR
DELTA FUNCTION
3 DIMENSIONS
LINEAR
DELTA FUNCTION
2 DIMENSIONS
1 DIMENSION
V
T



A — m = 1
B — m = 2, R_0=0.25
C — m = 2, R_0=0.05
D — m = 3, R_0=0.25
E — m = 3, R_0=0.05
V_2
T_1

absorption and infiltration for $m = 1, 2,$ and 3 with R_0 values 0.05 and 0.25 for the cases $m = 2, 3$.

Figure 20 shows the dependence of $V_2(\infty)$ on R_0 for $m = 2$ and 3 as indicated by Eqs. (158) and (164). The value $V_2(\infty) = 1$ for $m = 1$ is shown for comparison. Note that, in view of the definition of R_0 in Eq. (155), these curves can be interpreted as showing, for fixed r_0 and k, *the variation of the final infiltration rate with the capillary properties of the soil.*

We see that final infiltration rates for $m = 2$ and 3 depend in a complicated way on both the hydraulic and the capillary properties of the soil. It follows that final infiltration rates measured in the field by (necessarily finite) infiltrometers are *not*, as is often supposed, simply measures of the hydraulic properties of the soil. They are not directly related to (and are not necessarily well correlated with) the final infiltration rate for one-dimensional infiltration over large areas, which (for a uniform soil) depends only on hydraulic properties.

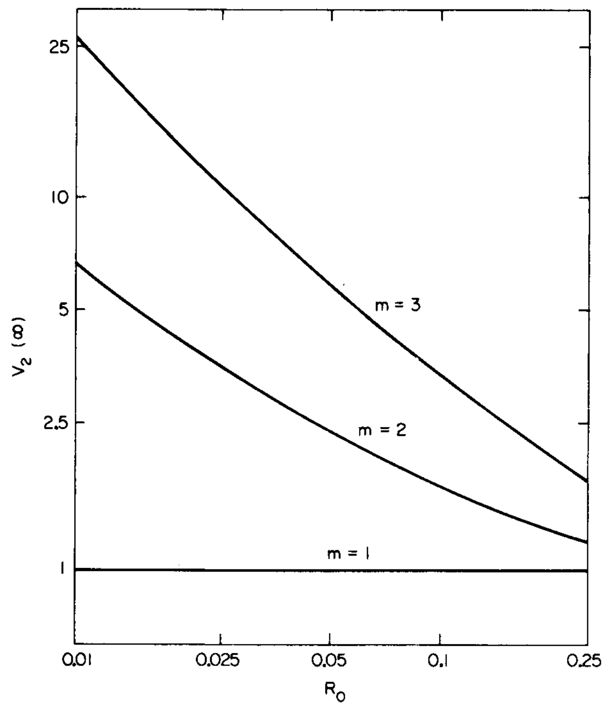


FIG. 20. The influence of geometry on final infiltration rate in the dimensionless form $V_2(\infty)$ as a function of R_0 and m. $V_2(\infty)$ for $m = 2$ and 3 are from Eqs. (158) and (164), respectively.

Symbols

Some symbols have been used in more than one way, but this should not lead to confusion. We indicate consistent convenient units where these are relevant.

- A — $(A_2 + K_0)$, coefficient of second term of series in Eq. (95) and of infiltration equation (195) (cm-sec^{-1})
- A_n — $\int_{\theta_0}^{\theta_1} \phi_n \, d\theta$ (cm-sec$^{-n/2}$)
- A_{*_n} — Coefficient of nth term of series in Eq. (64) (cm^n-sec$^{-n/2}$)
- C — Capillary and hydrostatic potential difference between supply surface and (putative) wet front in delta-function model, taken as positive (cm)
- c — A small numerical factor
- D — Moisture diffusivity (cm^2-sec^{-1})
- D_s — Value of D for $\theta = \theta_s$ (cm^2-sec^{-1})
- D_* — Diffusivity in linearized model (cm^2-sec^{-1})
- $E(\theta, x)$ — Coefficient of $\partial\theta/\partial x$ in horizontal one-dimensional flow equation in heterogeneous media (cm-sec^{-1})
- erfc — The complementary error function
- F — Function entering solution for one-dimensional absorption with constant flux condition (sec$^{-1/(n+2)}$)
- $F(\theta)$ — Function entering analytical solution for one-dimensional absorption (cm-sec$^{-1/2}$)
- $F(\theta, x)$ — Zeroth-order term in horizontal one-dimensional flow equation in heterogeneous media (sec^{-1})
- $f(\theta)$ — Any function of θ
- g — Acceleration due to gravity, ≈ 980 cm-sec^{-2}
- h — Relative humidity
- I — Dimensionless form of i defined in Eq. (136)
- I_1 — Dimensionless form of cumulative infiltration defined in Eq. (148)
- i — Cumulative absorption or infiltration (cm)
- $\mathbf{J}_0$ — Bessel function of the first kind of zero order
- K — Hydraulic conductivity (cm-sec^{-1})
- K_0 — Value of K for $\theta = \theta_0$ (cm-sec^{-1})
- K_1 — Value of K for $\theta = \theta_1$ (cm-sec^{-1})
- $\bar{K}$ — Value of K adopted for region between supply surface and (putative) wet front in delta-function model (cm-sec^{-1})
- K_{**} — Reduced hydraulic conductivity (cm^{-1}-sec^{-1})
- $\mathbf{K}_0$ — Modified Bessel function of the second kind of zero order
- k — Constant coefficient corresponding to $dK/d\theta$ in linearized model (cm-sec^{-1})
- m — Number of space dimensions
- n — Generally, a positive integer; in Sect. IX.A.1, the index of the power law dependence of D on $(\theta - \theta_0)$
- P — Pressure (dyne-cm^{-2})
- Q_2 — Two-dimensional source strength (cm^2-sec^{-1})
- Q_3 — Three-dimensional source strength (cm^3-sec^{-1})
- q — Total absorption rate from finite supply surface in three-dimensional systems (cm^2-sec^{-1})
- R — Dimensionless form of $(x^2+y^2)^{1/2}$ in three-dimensional systems, defined in Eq. (124)
- R_0 — Dimensionless form of r_0 defined in Eq. (155)
- R_* — Dimensionless form of r defined in Eq. (155)
- $\mathbf{R}$ — Gas constant for water vapor $\approx 4.615 \times 10^6$ erg-gm^{-1}-°K^{-1}
- r — Radius (cm)
- r_0 — Radius of supply surface (cm)
- r_*^{-1} — Mean curvative of supply surface (cm^{-1})
- S — Sorptivity (cm-sec$^{-1/2}$)
- $\mathscr{S}$ — Intrinsic sorptivity (cm$^{1/2}$)
- T — Dimensionless form of t defined in Eq. (136)
- T_1 — Dimensionless form of t defined in Eq. (148)

T Absolute temperature (°K)

t Time, normally with $t = 0$ at beginning of infiltration process (sec)

t_0 Arbitrary constant in Eq. (103) (sec)

t_{geom} Characteristic time based on capillary properties of medium and geometry of supply surface, defined by Eq. (69) (sec)

t_{grav} Characteristic time during which gravity becomes significant during infiltration, defined by Eq. (98) (sec)

U_0 Flow velocity at supply surface in processes subject to flux conditions (cm-sec^{-1})

U Vector flow velocity (cm-sec^{-1})

u Velocity of the "profile at infinity" in one-dimensional infiltration; see Eq. (106) (cm-sec^{-1})

UWR Ultimate wetted region; see Sect. XI

V Dimensionless form of v_0 defined in Eq. (136)

V_1 Dimensionless form of infiltration rate defined in Eq. (148)

V_2 Dimensionless form of $\bar{v}_0$ defined in Eq. (155)

V The volume of the medium

v Flow velocity (cm-sec^{-1})

v_0 Flow velocity at the supply surface; the absorption or infiltration rate based on unit area of supply surface (cm-sec^{-1})

$\bar{v}_0$ v_0 averaged over supply surface (cm-sec^{-1})

W The supply surface

w Dummy variable of integration

X Dimensionless form of $|x|$ in two-dimensional systems defined in Eq. (119)

x Horizontal Cartesian coordinate (cm)

Y_1 Coefficient in delta-function infiltration equation (209) (sec-cm^{-1})

$\mathbf{Y}_0$ Bessel function of the second kind of zero order

y Horizontal Cartesian coordinate (cm)

Z Dimensionless form of z defined in Eq. (119)

Z_1 Coefficient in delta-function infiltration equation (209) (cm)

z Vertical Cartesian coordinate; positive upward in Sect. II, positive downward in treatment of infiltration in later sections (cm)

α In Sect. II, the "wetting potential" of a volume element of the porosity of a medium (cm); also the constant defined by Eq. (112), which enters the representation of K as an exponential function of Ψ; see Eq. (113) (cm^{-1})

β In Sect. II, the "draining potential" of a volume element of the porosity of a medium (cm); also a constant entering the second of Eqs. (208)

γ Euler's constant $= 0.57722\ldots$

δ Dirac delta function

$\zeta(\theta)$ The ordinate of the "profile at infinity" in one-dimensional infiltration; see Eq. (107) (cm)

η A space coordinate (cm)

η_{wf} Distance from supply surface to wet-front in delta-function model (cm)

Θ Reduced form of θ defined in Eq. (199)

Θ_* Reduced form of Θ defined in Eq. (202)

θ Volumetric moisture content

θ_0 Initial value of θ

θ_1 Value of θ at the supply surface for $t \geqq 0$

θ_a Value of θ at which zero of ζ is taken

θ_s Value of θ corresponding to $\Psi = 0$

θ_U Asymptotic (large t) value of θ at $z = 0$ for one-dimensional infiltration with constant flux condition

θ_* $\int_{\theta_0}^{\theta} D\, d\theta$ (cm^2-sec^{-1})

ϑ θ-like variable used in linearized model

ϑ_2 Dimensionless form of θ_*, defined in Eq. (119)

ϑ_3 Dimensionless form of θ_*, defined in Eq. (124)
ϑ_* Transformed form of ϑ defined by Eq. (152)
κ Intrinsic permeability (cm^2)
λ Characteristic length scale of internal geometry of medium (cm)
μ Dynamic viscosity ($gm\text{-}cm^{-1}\text{-}sec^{-1}$)
ρ In Sect. II, the density of liquid water ≈ 1 $gm\text{-}cm^{-3}$; dimensionless radius ($= r/r_0$)
σ Surface tension ($dyne\text{-}cm^{-1}$)
τ Reduced form of t, defined in Eq. (48) ($sec\text{-}cm^{-2}$)
τ_1 Dimensionless form of t, defined in Eq. (199)
Φ Total potential (cm)
ϕ Similarity variable for one-dimensional absorption, defined by Eq. (33) ($cm\text{-}sec^{-1/2}$)
ϕ_n Coefficient of nth term of series in Eq. (83) ($cm\text{-}sec^{-n/2}$)
ϕ_{*n} Coefficient of $(n+1)$th term of series in Eq. (51) ($cm^n\text{-}sec^{-n/2}$)
χ Dimensionless form of x, defined in Eq. (199)
χ_* Reduced form of χ, defined in Eq. (202)
Ψ Moisture potential (cm)
Ψ_0 Initial value of Ψ (cm)
Ψ_1 Value of Ψ at the supply surface for $t \geqq 0$ (cm)
Ψ_e "Air-entry value" of moisture potential (cm)
Ψ_* $\int_{\Psi_0}^{\Psi} K\, d\Psi$ ($cm^2\text{-}sec^{-1}$)
Ψ_{**} Reduced moisture potential (cm^2)
ψ Angle defined in Eq. (155) (rad)
Ω Potential of external forces (cm)
$\omega(\vartheta)$ Function entering Eq. (147) ($cm\text{-}sec^{-1/2}$)
∇ Gradient of a scalar
$\nabla \cdot$ Divergence of a vector
∇^2 $\nabla \cdot \nabla$

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